



CHAPTER 8

Complex Integration: Cauchy's Theorem

8.1 Introduction

In the last few chapters our efforts to extend the idea of differentiation to complex mappings have been amply rewarded. By innocently attempting to generalize the real derivative we were quickly led to the amplitwist concept, and the subject then came to life with a character all its own. While many of the results cast familiar shadows onto the world of the reals, many did not, and striking indeed was the flavour of the arguments used to grasp them. The ability of z to freely roam the plane unleashed in us a degree of visual imagination that had to remain dormant so long as we could only watch the real number x forlornly pacing its one-dimensional prison.

This little hymn to the glory of the complex plane can be sung again in the context of integration, only louder. If differentiation breathed life into the subject, then integration could be said to give it its soul. Only after we have understood this soul will we be able to demonstrate such fundamental facts as the infinite differentiability of analytic mappings¹.

In ordinary calculus the symbol $\int_a^b$ has a clear meaning. However, if we wish to generalize this to $\mathbb{C}$ then the need for new ideas is immediately apparent, for how are we to *get* from a to b ? In $\mathbb{R}$ there was only one way, but a and b are now points in the plane, so we must specify some connecting path (called a *contour*) “along which to integrate”. It is then natural to ask whether the value of the integral depends upon the choice of this contour.

In general the value of the integral *will* depend on the route chosen. For example, we will shortly meet an integral of a complex mapping that yields, when evaluated for a closed contour, the area enclosed by the contour—a flagrant dependence of

¹ Since the 1960's it has actually become possible to do such things without integration, thanks to pioneering work by G. T. Whyburn, and others. Nevertheless, integration still appears to provide the simplest approach.

value on contour. It should be made clear from the outset that while differentiation only made sense for the strictly limited set of analytic functions, this is *not* the case for integration. Indeed, the example just cited involves the integration of a non-analytic function.

The principal aim of this chapter (beyond the mere construction of an integral calculus) will be the discovery of conditions under which the value of an integral does *not* depend on the choice of contour. One such result is an analogue of the Fundamental Theorem of real analysis, and in deference to that subject it bears the same name. However, in the complex realm this is actually a misnomer, for there exists a still deeper result which has no counterpart in the world of the reals. It is called *Cauchy's Theorem*.

As we have said, it is not only possible, but sometimes useful to integrate non-analytic functions. However, it should come as no surprise to learn that new phenomena arise if we concentrate on the integrals of mappings that *are* analytic. Cauchy's Theorem is the essence of these new phenomena. Essentially it says that any two integrals from a to b will agree, provided that the mapping is *analytic everywhere in the region lying between the two contours*. Almost all the fundamental results of the subject (including some already stated) flow from this single horn of plenty.

8.2 The Real Integral

8.2.1 The Riemann Sum

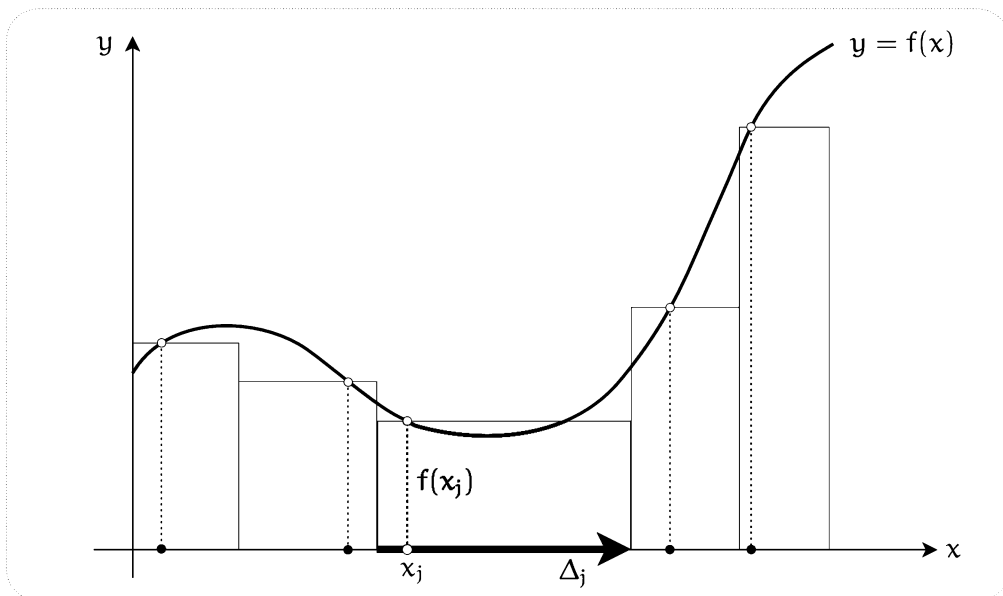
As we did with differentiation, we begin by re-examining the more familiar idea of integrating a real function. The historical origin of this process, and still the principal means of visualizing it, is the problem of evaluating the area under the graph of a function.

We first approximate the sought-after area with rectangles. See [8.1]. Dividing the interval of integration into n line-segments Δ_j (the bases of the rectangles), we randomly select one point x_j from each segment, and take the height of the corresponding rectangle to be the height of the curve above the point, namely, $f(x_j)$. The area of each rectangle is then $f(x_j) \Delta_j$, and thus the total rectangular approximation to the area under f is

$$R \equiv \sum_{j=1}^n f(x_j) \Delta_j . \quad (8.1)$$

The quantity R is called a *Riemann sum*. Finally, by simultaneously letting n tend to infinity while each Δ_j shrinks to nothing, R will tend to the desired area.

In [8.1] we could afford to be indifferent to the precise choice of x_j within each Δ_j because we had our eye on this final limiting process. As each Δ_j shrinks, the



[8.1] A Riemann Sum $R \equiv \sum_{j=1}^n f(x_j) \Delta_j$ approximates the area under a curve with rectangles, the exact answer being obtained in the limit that n goes to infinity and each Δ_j goes to zero.

freedom in the choice of x_j becomes more and more limited, and the influence of the choice on the area of the rectangle likewise diminishes. However, if we are unwilling or unable to actually carry out the limiting process, then, as we shall now see, we can ill afford to be so blasé in our choice of x_j .

You probably dimly remember some professor showing you (8.1) before, and perhaps you even evaluated a couple of examples by means of it. However, this was no doubt quickly forgotten once you set eyes upon the Fundamental Theorem of Calculus. In order to integrate x^4 , why bother with taking the limit of some complicated series when we know that the answer must be that function which differentiates to x^4 , namely, $\frac{1}{5}x^5$?

The Fundamental Theorem *is* a wonderful thing, but one must remember that many quite ordinary functions simply do not possess an antiderivative that is expressible in terms of elementary functions. To take a simple example, the Normal Distribution of statistics requires a knowledge of the area under the curve e^{-x^2} , and this can only be computed *numerically*, perhaps via a Riemann sum.

When doing a numerical calculation with (8.1), it would require an infinite amount of time to find even a single area with perfect precision. It is therefore important to be able to obtain good approximations to $\lim_{n \rightarrow \infty} R$ while using only a finite value of n . Several such methods exist: Simpson's rule and the Trapezoidal

rule, to name just two that may be familiar. Since the Trapezoidal rule will most readily lend itself to complex generalization, we will now review it.

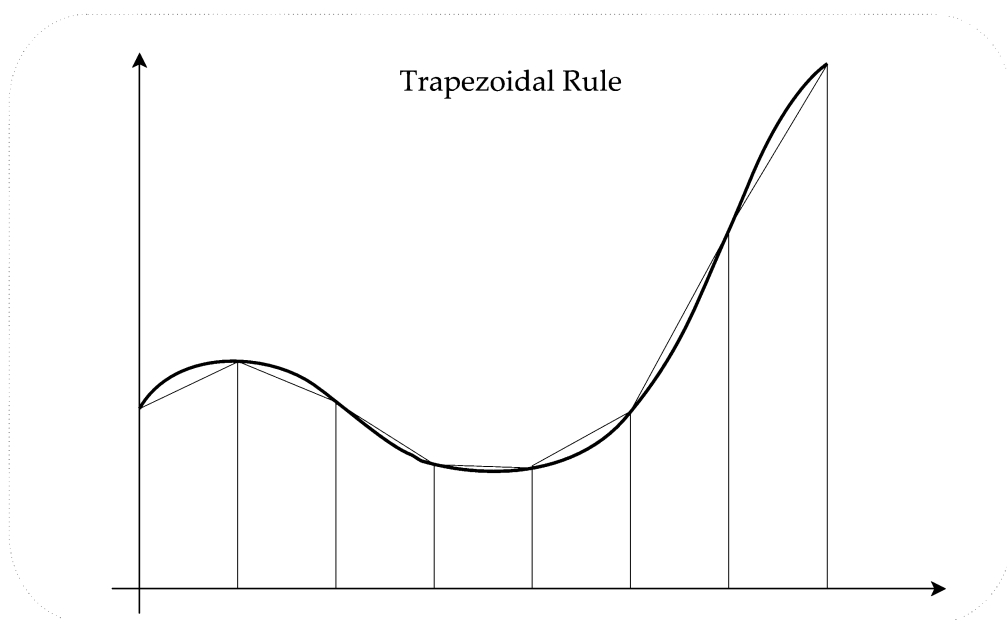
8.2.2 The Trapezoidal Rule

As the name suggests, we now use trapezoids instead of rectangles to approximate the area. Though not strictly necessary, it is convenient to make all the Δ_j the same length. See [8.2].

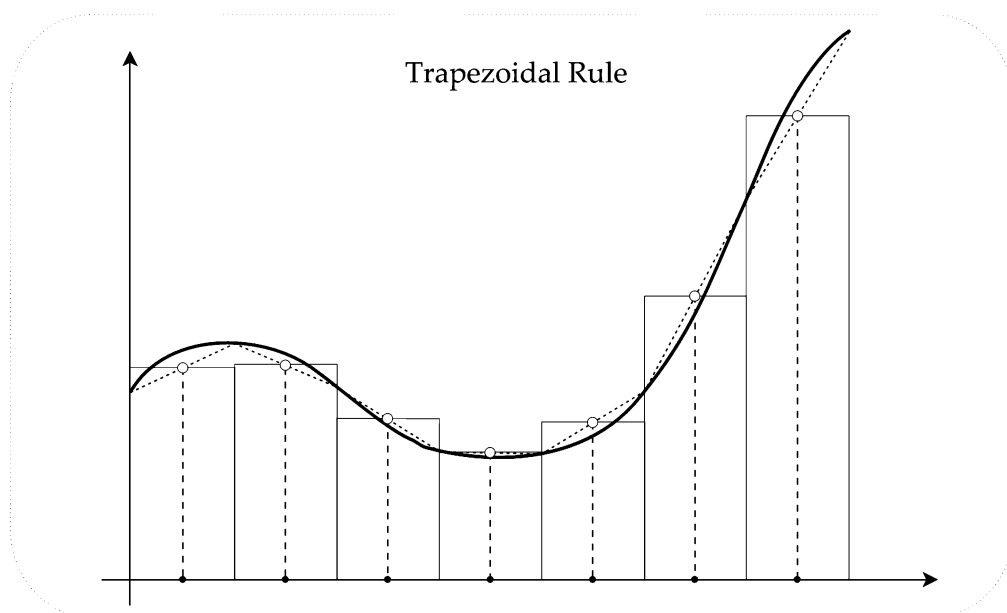
It is clear from the figure that even a very modest value of n will yield a quite accurate estimate. Since [8.2] is not of the same form as [8.1], the associated Trapezoidal Formula (which we won't bother to state) is not quite of the type (8.1). Nevertheless, if we wish to continue to use (8.1), it is not hard to find a Riemann sum that closely mimics the trapezoidal sum, and hence which retains the latter's accuracy.

First note that the shaded trapezoidal estimate shown in [8.2] is identical to the rectangular one in [8.3], in which we have taken the height of each rectangle to be the height of the chord at the midpoint of Δ_j . Finally, to recover a Riemann sum, we can replace the height of the chord by the height of the curve at that point. See [8.4].

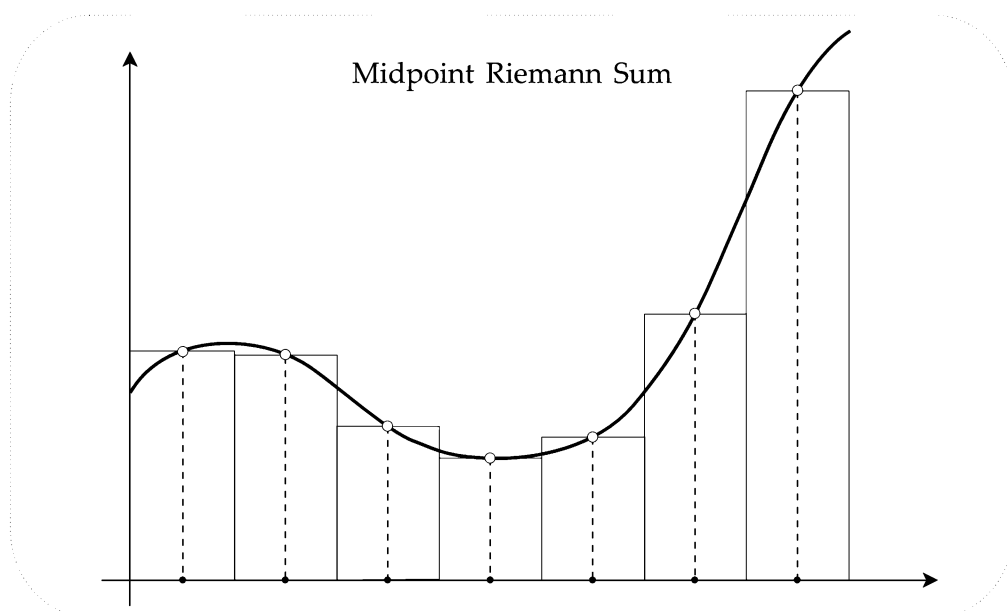
In other words, the Riemann sum (8.1) will yield an accurate approximation to the integral, using only a modest value of n , provided that we choose each x_j to be at the midpoint of its Δ_j . We will call this the *Midpoint Riemann Sum*, and write it as R_M .



[8.2] The Trapezoidal Rule replaces the rectangles with trapezoids, and clearly yields an accurate approximation to the integral while using only a modest value of n .



[8.3] *The exact same area as the previous figure, but now represented as rectangles whose heights are given by the midpoints of the illustrated chords.*



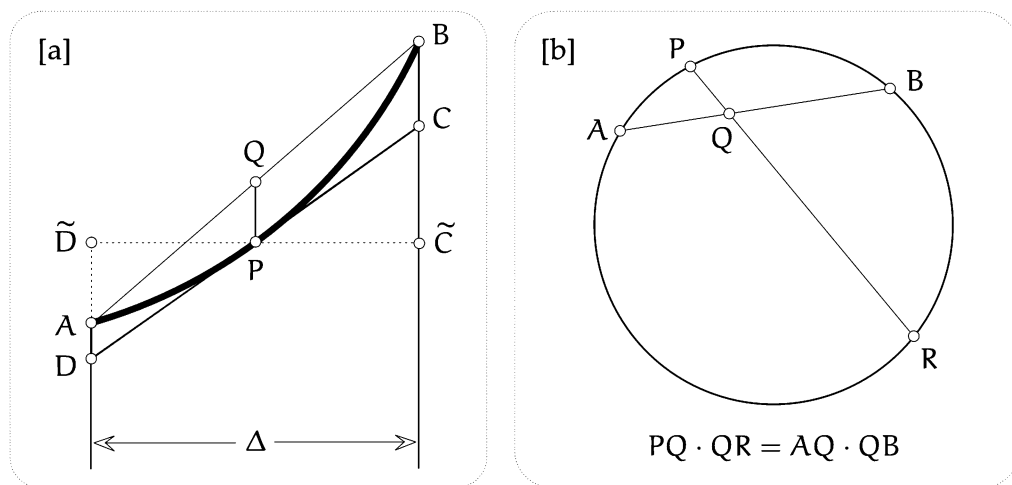
[8.4] *The Midpoint Riemann Sum, R_M , mimics the previous figure, but now uses the height of the graph at each midpoint, instead of the height of the chord. As an approximation to the integral, a typical Riemann sum that uses the height at a random point has a total error that dies away as Δ , but we will show that the Midpoint Riemann Sum is much more accurate, with an error that dies away as Δ^2 .*

8.2.3 Geometric Estimation of Errors

We have said that using midpoints in (8.1) will yield accurate results, but how accurate is “accurate”? First reconsider the case where the x_j were chosen randomly, and suppose that all the Δ_j 's have the same length. Re-examination of [8.1] reveals that the difference between the actual area lying above each Δ_j , and the area of the approximating rectangle, will be of order Δ^2 . Since the total number of rectangles is of order $1/\Delta$, it follows that the total error will be of order Δ , and thus, as claimed, it will die away as n increases and Δ shrinks. We will now show that using R_M , or the almost equivalent Trapezoidal rule, produces a much smaller total error—in fact an error that dies away as the *square* of Δ .

This standard result on the decay of the error can be found in many advanced calculus books, but rather than repeat the standard calculation, we will supply a novel geometric² account. Figure [8.5a] shows a magnified view of the top of one of the rectangles used in R_M . Shown are the chord AB bounding the trapezoid used in [8.2], and the line-segment $\tilde{D}\tilde{C}$ bounding the rectangle used in R_M . Notice that P and Q (the midpoints of these line-segments) will lie directly above the point x_j that is being used in R_M .

Visually, it is easy to compare the area under AB [the Trapezoidal rule] with the actual area under the curve, but the same cannot be said of the area under



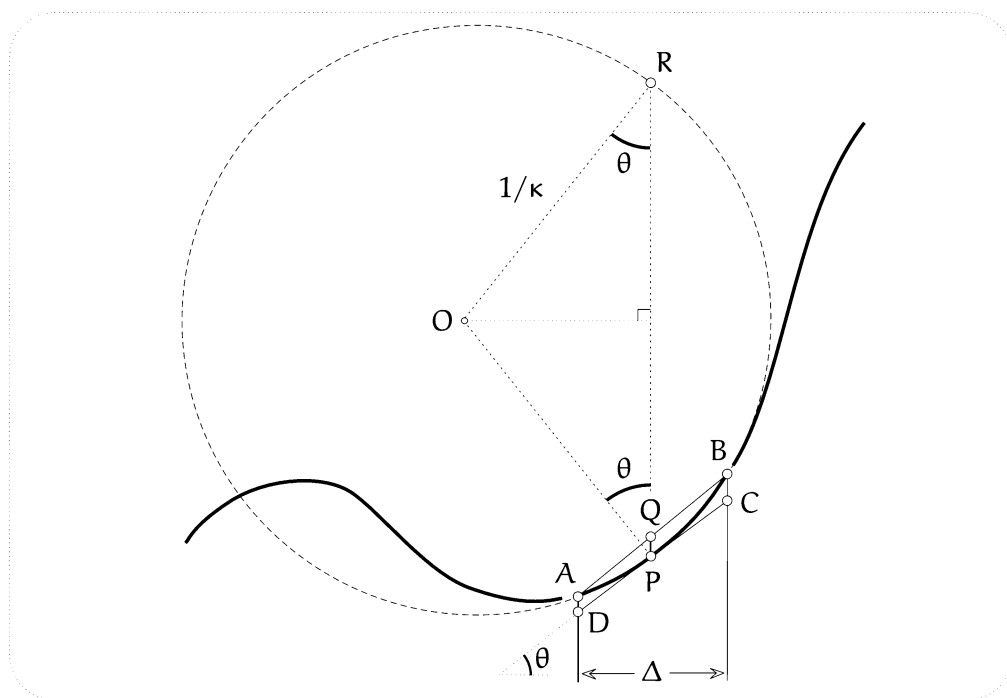
[8.5] [a] Start of geometric determination of the accuracy of the Midpoint Riemann Sum.

The area of the rectangle occurring in R_M is the area under $\tilde{D}\tilde{C}$, but this is the same as the area under the tangent, DC , so the error is the shaded region between DC and the curve. This error is certainly less than the area of $ABCD$, which is $(PQ) \cdot \Delta$. **[b]** As the chord PQR revolves about the fixed point Q , the product $PQ \cdot QR$ remains constant.

² While many of the arguments in this book were merely inspired by Newton's mode of thought in the *Principia*, we have here an example that is very close to his actual methods.

$\tilde{D}\tilde{C}$ [the R_M rule]. However, note that if we rotate $\tilde{D}\tilde{C}$ about P (keeping the ends glued to the verticals) until it becomes *tangent* at DC , the area beneath it will remain constant [why?]. Thus we are instead free to visualize each term of R_M as being the area lying beneath a tangent such as DC . It is now clear that the actual area lies between the two values furnished by AB and DC , and that the error induced by using either rule cannot exceed the area of the small quadrilateral $ABCD$, namely [exercise], $(PQ) \cdot \Delta$. In order to find this area we will use the elementary property of circles that is illustrated in [8.5b]: *As the chord PQR revolves about the fixed point Q , the product $PQ \cdot QR$ remains constant.*

Over a sufficiently tiny distance we can consider any segment of curve to be interchangeable with its tangent. However, over somewhat larger distances (or if we simply require greater accuracy) we must instead replace it by a segment of its *circle of curvature*, that is, the circle whose curvature κ agrees with that of the curve at the point in question. In [8.6] we have drawn this circle for the segment at P . The above result now informs us that



[8.6] Conclusion of geometric determination of the accuracy of the Midpoint Riemann

Sum. Let us approximate the curve in the vicinity of P with its circle of curvature, of radius $(1/\kappa)$. By virtue of the previous figure, $PQ \cdot QR = (AQ)^2$. But in the limit of vanishing Δ , we have $AQ \asymp DP$ and $QR \asymp PR$, so $PQ = (AQ)^2/QR \asymp (DP)^2/PR$. If θ is the angle of the tangent at P , then $DP = \frac{1}{2}\Delta \sec \theta$, and $PR = (2/\kappa) \cos \theta$. Therefore, $\text{area}(ABCD) = PQ \cdot \Delta \asymp (\frac{1}{8}\kappa \sec^3 \theta) \Delta^3$.

$$PQ \cdot QR = (AQ)^2. \quad (8.2)$$

As Δ shrinks, both (AQ/DP) and (QR/PR) tend to unity, so in this limit we may substitute DP for AQ and PR for QR . But if the tangent at P makes an angle θ with the horizontal (in which case OP makes angle θ with the vertical) then $DP = \frac{1}{2}\Delta \sec \theta$, and $PR = (2/\kappa) \cos \theta$. Substituting these into (8.2), we obtain the result

$$\text{area}(ABCD) = PQ \cdot \Delta \asymp \left(\frac{1}{8}\kappa \sec^3 \theta\right) \Delta^3. \quad (8.3)$$

If M denotes the maximum of $\left(\frac{1}{8}\kappa \sec^3 \theta\right)$ over the integration range (which we take to be of length L) then each such error will be less than $M \Delta^3$. Since the number of these error terms is (L/Δ) , we conclude that

$$\text{total error} < (LM) \Delta^2,$$

and this indeed dies away in the manner originally claimed. [At this point you may care to look at Ex. 1]

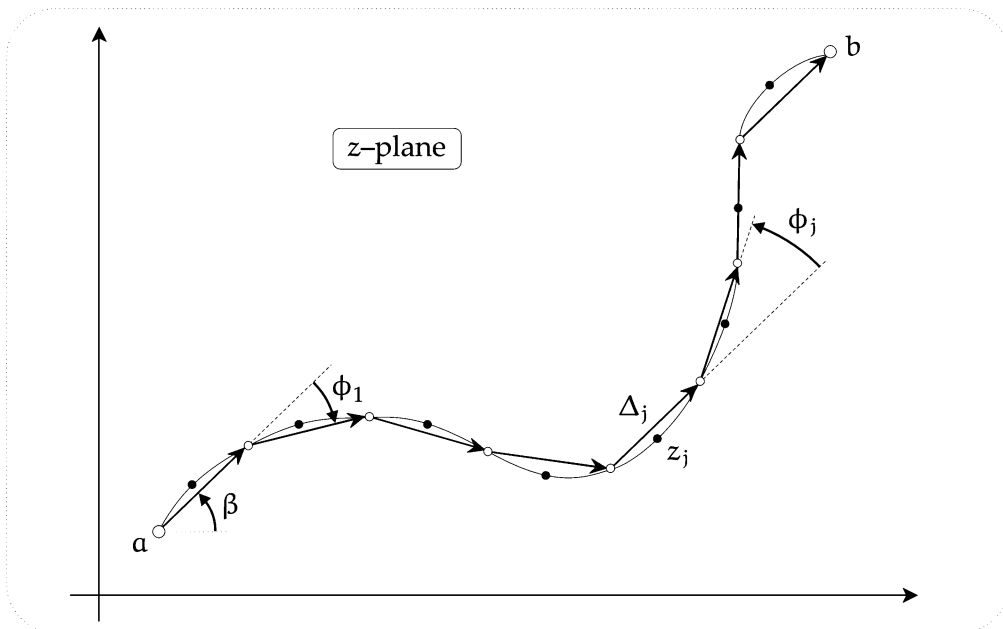
Because the order of the induced error is the same for both R_M and the Trapezoidal rule, we will tend not to distinguish between them when it comes to their complex generalizations. This said, there remains one curious pedagogical point still to be made.

Figure [8.5a] makes it clear that curves deviate less from their tangents than from their chords, and thus one would anticipate that while the order of the error is the same for both rules, R_M would actually yield the more accurate value of the two. This is indeed the case, and in fact [see Ex. 2] one can show that it is *twice* as accurate. In addition to this accuracy, R_M is, if anything, easier to remember and use than the Trapezoidal formula. It is therefore doubly puzzling that the Trapezoidal formula is taught in every introductory calculus course, while it appears that the midpoint Riemann sum R_M is seldom even mentioned.

8.3 The Complex Integral

8.3.1 Complex Riemann Sums

In the case of real integration we began with a clear geometric objective ("Find the area!") and then invented the integral as a means to this end. In the complex case we will reverse this process, that is, we will first blindly attempt to generalize real integrals (via Riemann Sums) and only afterwards will we ask ourselves what we have created. First, in this chapter, we will find one way of picturing an integral as a single complex number; then, in Chapter 11, we will use an entirely different point of view to see that, separately, the real and the imaginary parts of an integral each possess a vivid geometric (and physical) significance. But to guess the relevant geometric entities in advance, and then to invent the complex integral as the appropriate tool with which to find them, would require a prodigious leap of



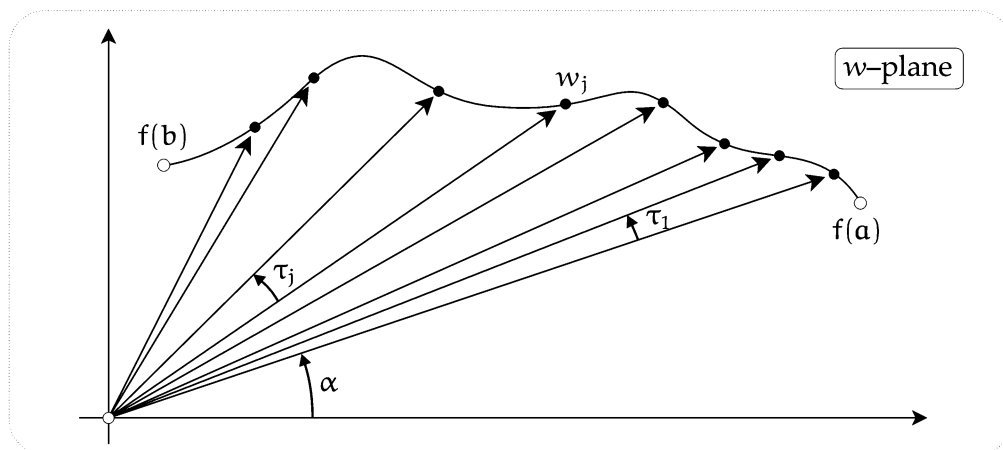
[8.7] Complex Riemann Sums. To approximate $\int_K f(z) dz$, we approximate K with many small complex chords, Δ_j , evaluate $f(z)$ at an arbitrary point z_j within each segment of K , then evaluate the sum, $\sum f(z_j) \Delta_j$. Given the established superior accuracy of the Midpoint Riemann Sum (R_M) in the real case, here we have immediately chosen each z_j to be the midpoint of each segment. As we shall explain, the Riemann sum can be visualized more easily if we choose all the Δ_j 's to have the same length. [The turning angles ϕ_j illustrated here are for later use.]

imagination—one that historically never took place. A moment's thought reveals that this is similar to the case of differentiation, for there we began with the slope concept, and through an initially blind process of extrapolation we arrived at the very different (but no less intuitive) idea of the amplitwist.

Consider [8.7]. In order to integrate a complex mapping $f(z)$ between the points a and b , we have specified a connecting curve (*contour*) along which to perform the integration. This curve (call it K) now plays the role of the interval of integration, and just as in [8.1], we break it down into small steps Δ_j , which we may conveniently choose to be of equal length. The difference between this and [8.1] is that now the steps are not all in the same direction.

In order to construct a Riemann sum, we randomly pick one point z_j from each little segment of K , and then we form the sum of the products $f(z_j) \Delta_j$. Finally, as we increase their number and decrease their lengths, the Δ_j will follow K ever more perfectly, and the Riemann sum will tend to a limiting value (provided only that the mapping is continuous) that serves as our definition of the complex integral, written

$$\int_K f(z) dz .$$



[8.8] The values of the function along the contour of integration: $w_j = f(z_j)$. [The angles τ_j between w_j and w_{j+1} are for later use.]

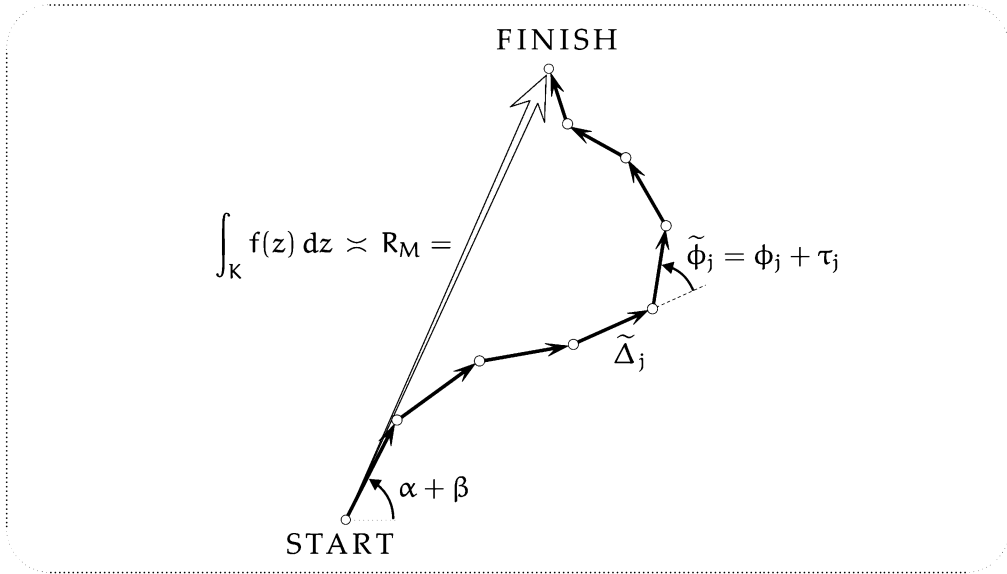
Just as in the real case, we may obtain an accurate estimate of the integral without passing to the limit, simply by choosing the z_j to be at the *midpoints* of the segments of K , rather than at random points. In fact this is the choice that we have illustrated in [8.7]. Once again, this especially accurate Riemann sum will be denoted R_M .

To begin to understand the geometry of R_M , consider [8.8]. This shows the image of K under the mapping $z \mapsto w = f(z)$, and in particular the image w_j of the z_j that was singled out in [8.7]. The corresponding term of R_M is then $\tilde{\Delta}_j \equiv w_j \Delta_j$, and we will choose to think of this as the arrow that results when w_j “acts on” Δ_j , expanding it by $|w_j|$ and rotating it by $\arg(w_j)$.

Having obtained each $\tilde{\Delta}_j$ in this manner, we go on to join all these little arrows together (tail to tip), as in [8.9]. The value of R_M , and hence the approximate value of the integral, is then the connecting complex number between the start and the finish³. Notice that since the answer is a *connecting* arrow, the point at which we begin drawing R_M is irrelevant.

While [8.9] is intended primarily to convey the general idea, it is in fact a faithful evaluation of the specific R_M corresponding to [8.7] and [8.8], and you may now begin to convince yourself of this. This is perhaps most easily achieved by concentrating on the lengths of the $\tilde{\Delta}_j$ separately from their angles. As w traces out the image curve in [8.8], its length diminishes, and this produces a corresponding shrinking of the $\tilde{\Delta}_j$ in [8.9]. Likewise, the increasing angle of w results in progressively greater rotations of the Δ_j .

³ The great physicist Richard Feynman used a similar kind of picture to explain his quantum-mechanical “path integrals”, which are also complex, though they differ from contour integrals. See Feynman (1985).



[8.9] Visualizing the Midpoint Riemann Sum, R_M . The angle of each term $\tilde{\Delta}_j \equiv w_j \Delta_j$ in $R_M = \sum \tilde{\Delta}_j$ is the sum of the angles of w_j and Δ_j , so the illustrated turning angle $\tilde{\phi}_j$ between successive steps along the Riemann sum is therefore the sum of the turning angles ϕ_j and τ_j in the previous two figures: $\tilde{\phi}_j = \phi_j + \tau_j$.

8.3.2 A Visual Technique

While it is not strictly necessary to choose the same length for all the Δ_j , the benefit of this choice is probably clear: the lengths of the $\tilde{\Delta}_j$ are simply proportional to $|w_j|$, and it is therefore easy to follow the evolution of $|\tilde{\Delta}_j|$ by eye. On the other hand, it is not so easy to visually follow the evolution of the *angle* of $\tilde{\Delta}_j$.

As we travel along the Δ_j in [8.7], we pass through a sequence of sharp bends. The turning angle at a typical bend is drawn in [8.7], and is denoted ϕ_j . What will be the turning angle $\tilde{\phi}_j$ at the corresponding bend of the Riemann sum? If, for example, w_{i+1} pointed in the same direction as w_j , then Δ_{i+1} and Δ_j would both suffer the same rotation, and the turning angle $\tilde{\phi}_j$ of the Riemann sum would equal the original turning angle ϕ_j . More generally, if the angle of w increases by τ_j (see [8.8]), then the turning angle will also increase by τ_j . Thus,

$$\tilde{\phi}_j = \phi_j + \tau_j. \quad (8.4)$$

This simple observation helps to reduce the difficulty of visualizing R_M . It is no longer necessary to look at the angle of each w_j (which may be large and hard to gauge by eye) and to try and imagine the direction of the rotated Δ . In fact we need now only do this *once*, to find $\tilde{\Delta}_1$, thereby ensuring that R_M heads off in the correct initial direction. Thereafter, each successive $\tilde{\Delta}$ is laid down at an angle $\tilde{\phi}$ to its predecessor, and these $\tilde{\phi}_j$ may be readily estimated by eye, using (8.4).

Let us spell this out in detail with reference to the concrete example furnished by [8.7] and [8.8]. In [8.9], we get R_M started in the right direction by rotating Δ_1 by α , thereby obtaining $\tilde{\Delta}_1$ which points at angle $\alpha + \beta$. We can now draw the rest of R_M using only (8.4). To lay down the next $\tilde{\Delta}$ we need to know $\tilde{\phi}_1 = \phi_1 + \tau_1$. The small positive τ_1 clearly kills off just a fraction of the negative ϕ_1 , resulting in a slightly smaller negative bend in R_M . Much the same happens when we lay down $\tilde{\Delta}_2$. The angle ϕ_3 at the next bend is positive, and it is therefore increased by τ_3 , which itself is about twice as big as τ_1 and τ_2 were. You should now be in a position to follow the rest of R_M 's progress in far greater detail than you could before.

Although the above idea will shortly prove its worth on a theoretical level, it is clearly not terribly practical. However, in Chapter 11 we will use an entirely different approach to obtain a second, less strenuous, means of visualizing complex integrals. We will thereby make a double fallacy of an assertion that is to be found in most texts—assuming they even consider it worthy of note!—namely, that complex integrals possess *no* geometric interpretation. Perhaps the mere frequency with which this myth has been reiterated goes some way to explaining how it has acquired the status of fact.

8.3.3 A Useful Inequality

In figure [8.9] it is clear that if we were to straighten out all the bends in R_M then it would get longer. Furthermore, the length of the straightened version would just be the sum of the $|\tilde{\Delta}_j|$. Thus

$$|R_M| \leq \sum |w_j| \cdot |\Delta_j|,$$

with equality if⁴ and only if all $\tilde{\phi}_j = 0$. If M denotes the maximum distance from the origin to the image curve in [8.8], it follows that

$$|R_M| \leq M \sum |\Delta_j|.$$

But the sum on the right is just the length of the polygonal approximation to K , and hence it cannot exceed the actual length of K . Passing to the limit where R_M becomes the integral, we deduce that

$$\left| \int_K f(z) dz \right| \leq M \cdot (\text{length of } K). \quad (8.5)$$

For example, if $f(z) = (1/\bar{z})^2$ and K is the circle $|z| = r$, then (8.5) implies that $|\int_K f(z) dz| \leq (2\pi/r)$. In particular this implies that $\lim_{r \rightarrow \infty} \int_K f(z) dz = 0$. This is a typical (albeit simplistic) application of (8.5): quite often one wishes to demonstrate

⁴ The method of visualizing complex integrals in Chapter 11 will enable us to express this condition for equality in a particularly simple form. See Ex. 6, p. 574.

the ultimate vanishing of an integral as K evolves through some family, such as circles of increasing radius. Without knowing the exact value of any of the integrals, (8.5) shows that it is sufficient to demonstrate that the maximum size of $f(z)$ on K dies away faster than the length of K grows.

8.3.4 Rules of Integration

Because the complex integral has been defined in complete analogy with the real one, it follows that the former will inherit many of the properties of the latter. We now list some of these shared properties:

$$\begin{aligned}\int_K c f(z) dz &= c \int_K f(z) dz \\ \int_K [f(z) + g(z)] dz &= \int_K f(z) dz + \int_K g(z) dz \\ \int_{K+L} f(z) dz &= \int_K f(z) dz + \int_L f(z) dz \\ \int_{-K} f(z) dz &= - \int_K f(z) dz.\end{aligned}$$

The meaning of the first two equations is self-evident, but the last two require some clarification.

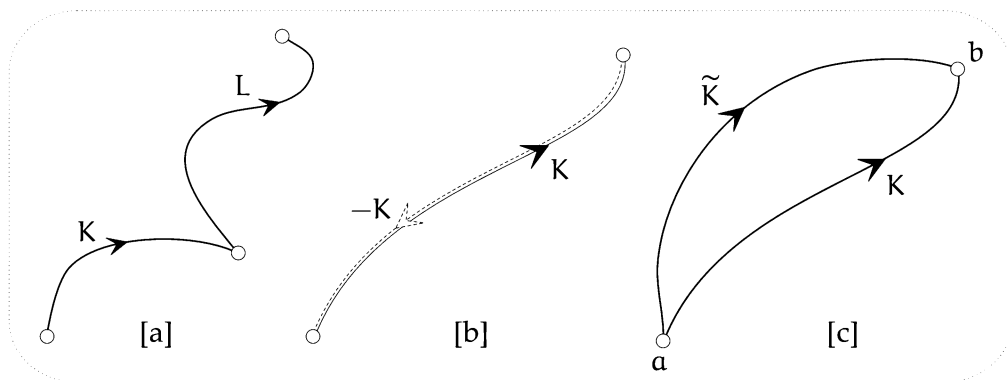
If L begins where K left off (see [8.10a]), then to integrate along $K + L$ means to integrate along K and then to continue integrating along L , and the resulting integral is then just the sum of the two separate integrals. Notice that the contour is allowed to have a kink in it. In fact the definition of “contour” merely requires that the number of such kinks not be infinite.

Lastly, the fourth rule is analogous to swapping the limits on a real integral, for $-K$ is defined to be the same as K , but traversed in the opposite direction (see [8.10b]).

However familiar you may be with these rules in real calculus, and however readily they may lend themselves to complex generalization, we would nevertheless urge you to make a new and separate peace with each of these results, preferably in terms of pictures such as [8.7], [8.8], and [8.9].

Recall from the Introduction that our main objective is the discovery of conditions under which an integral between two points in the plane does not depend on the connecting route chosen. The last two rules above may be used to recast this problem into a neater form. Suppose that the two paths K and $\tilde{K}$ in [8.10c] both yield the same value for the integral between a and b . It follows that

$$0 = \int_K f(z) dz - \int_{\tilde{K}} f(z) dz = \int_K f(z) dz + \int_{-\tilde{K}} f(z) dz = \int_{K-\tilde{K}} f(z) dz.$$



[8.10] [a] If L begins where K left off, then to integrate along $K + L$ means to integrate along K and then to continue integrating along L , and the resulting integral is then just the sum of the two separate integrals. [b] Changing K to $-K$ reverses the sign of the integral, for $-K$ is defined to be the same curve as K , but traversed in the opposite direction. [c] If the integral from a to b along K equals the integral along an alternative route $\tilde{K}$, then the integral along the closed loop $(K - \tilde{K}) = (K \text{ followed by } -\tilde{K})$ must be zero. Conversely, if the integral vanishes for all closed loops then all curves between a and b will yield the same value for the integral. In brief: **Path independence is equivalent to vanishing loop integrals.**

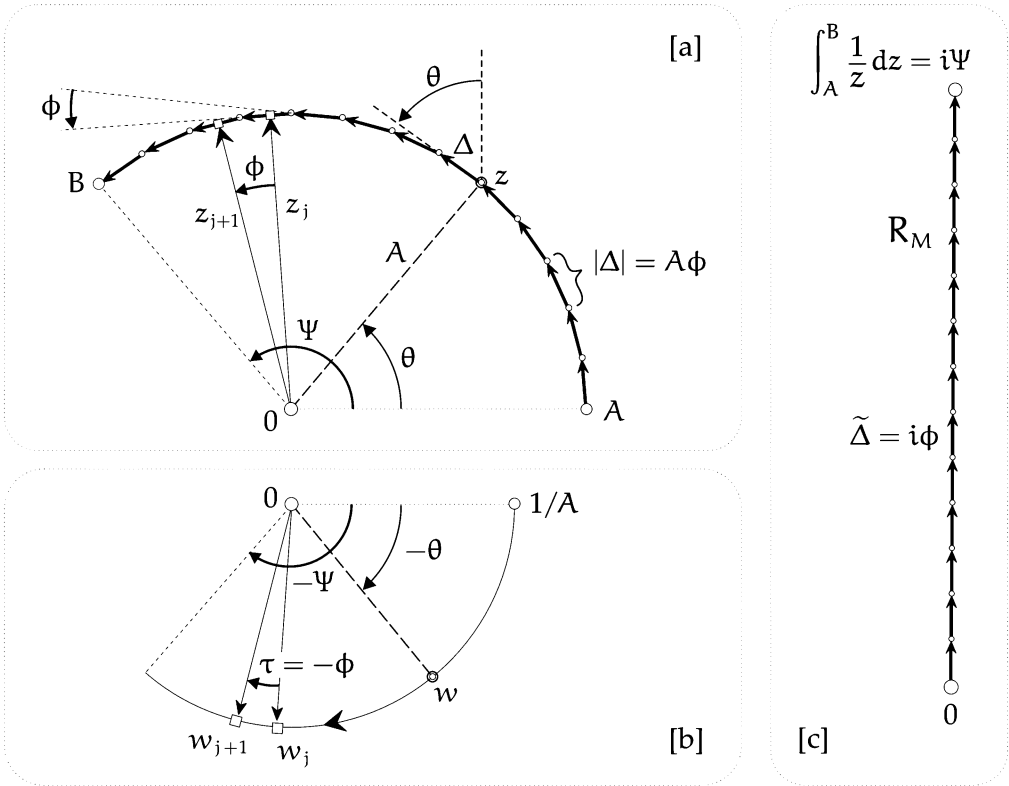
Thus equality of the two integrals is equivalent to the vanishing of the integral taken along the closed loop $(K - \tilde{K}) = (K \text{ followed by } -\tilde{K})$. Conversely, if the integral vanishes for all closed loops then all curves between a and b will yield the same value for the integral. In brief: *path independence is equivalent to vanishing loop integrals.* The centrepiece of complex analysis is the link between this phenomenon and analyticity. Cauchy's Theorem consists in recognizing that the vanishing of loop integrals is the nonlocal manifestation of a local property of the mapping, namely, that it is an amplitwist everywhere inside the loop.

8.4 Complex Inversion

8.4.1 A Circular Arc

Probably the single most important integral in all of complex analysis is that of the complex inversion mapping, $z \mapsto 1/z$. While the truth of this assertion will only emerge gradually, this is the reason for the great attention we will now lavish on this particular example.

We begin with the simplest case, namely, where the path of integration K is an arc of the origin-centred circle of radius A (see [8.11a]). As in [8.7], we divide this path into small (ultimately vanishing) steps of equal length. The turning angles ϕ_j clearly all have the same value, say ϕ . Since the angle that each Δ subtends at the



[8.11] The integral of $(1/z)$ along a circle. Since Δ_1 and w_1 are ultimately vertical and horizontal respectively, it follows that R_M (in [c]) initially heads off in the vertical direction. But since the turning angle τ of $w_j = (1/z_j)$ in [b] is the opposite of the turning angle ϕ of Δ in [a], $\tilde{\phi} = \phi + \tau = 0$. In other words, R_M has no bends, and so it continues on in the imaginary direction. We wanted to illustrate the turning-angle approach for later use, but here is a simpler explanation. Since the Δ_j located at angle θ on the circle itself points at θ to the vertical (as illustrated) and has length $A\phi$, we see that multiplying it by $w = (1/A)e^{-i\theta}$ rotates it by $-\theta$, producing a vertical $\tilde{\Delta} = w\Delta = i\phi$.

origin is also given by ϕ , it follows that $|\Delta| = A\phi$. As z travels round the circle its image $w = 1/z$ travels round a circle of radius $1/A$ in the opposite direction (see [8.11b]), and thus w shrinks each Δ to produce a $\tilde{\Delta}$ of length ϕ .

Since Δ_1 and w_1 are ultimately vertical and horizontal respectively, it follows that R_M (which we choose to begin drawing at the origin in [8.11c]) initially heads off in a vertical direction. But now we observe that $\tau = -\phi$, and consequently that $\tilde{\phi} = 0$. In other words R_M has no bends, and so it continues on in the imaginary direction⁵. Thus, irrespective of the radius, the integral equals i times the total angle Ψ through

⁵ We have used the turning angle idea in order to make the subsequent generalization to other powers of z straightforward, but there is actually no need for it in the present case. Since the Δ_j located at angle θ on the circle itself points at θ to the vertical and has length $A\phi$, we see that multiplying it by $w = (1/A)e^{-i\theta}$ rotates it by $-\theta$, producing a vertical $\tilde{\Delta} = w\Delta$ of length ϕ .

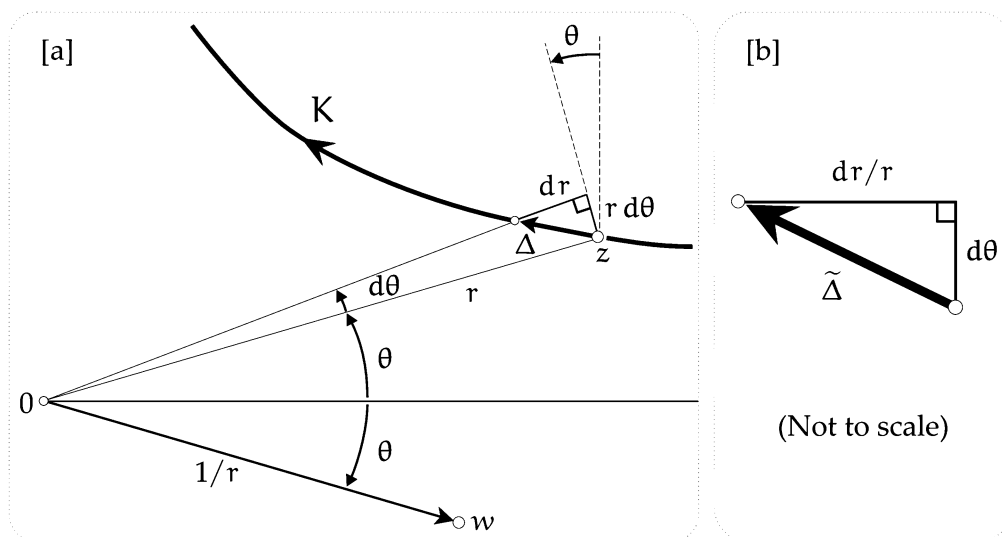
which z turned on its journey along K . Convince yourself that this formulation of the result remains valid even if K begins at a random point of the circle instead of on the real axis.

In particular, and of crucial importance, is the case where z continues all the way round the circle to form a closed loop. The value of the integral is then $2\pi i$. The alert reader will immediately be perplexed by this result. Why? Because it appears to fly in the face of Cauchy's Theorem. We have previously demonstrated geometrically that complex inversion is analytic, so how can its loop integral fail to vanish?! The resolution lies in the fact that Cauchy's Theorem requires that the mapping be analytic *everywhere* inside the loop. But our loop encloses the origin, and just at this one point the analyticity of complex inversion breaks down.

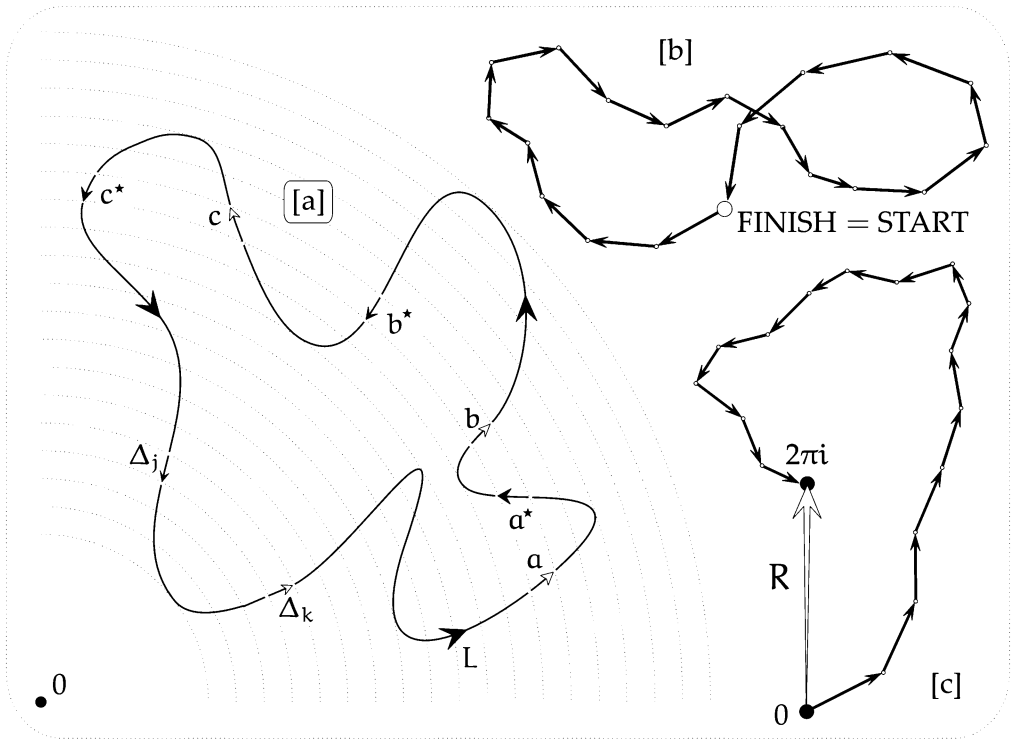
8.4.2 General Loops

The above discussion not only explains why our loop integral failed to vanish, but it also leads us to anticipate that if the loop does *not* enclose the origin, then the integral *will* vanish. We will now show that this is the case, thereby lending some credence to Cauchy's Theorem.

The ease with which we were able to evaluate the integral in [8.11] resulted from the fact that the Δ_j were all orthogonal to the z_j . Figure [8.12 a] shows a more typical Δ possessing a radial component in addition to a transverse one. As you see, Δ can be decomposed into a transverse component $r d\theta$ making an angle θ with the vertical, and an orthogonal radial component dr . To obtain the corresponding piece $\tilde{\Delta}$ of R (see [8.12b]) we multiply by w , thereby rotating these components into



[8.12] The integral of $(1/z)$ along a general contour. [a] The movement Δ along a general contour can be broken down into a radial component dr and an orthogonal component $r d\theta$ making angle θ with the vertical. [b] Multiplication by $w = (1/z)$ therefore yields $\tilde{\Delta} = w\Delta = (dr/r) + id\theta$.



[8.13] The integral of $(1/z)$ along a closed loop. [a] Instead of choosing Δ_j 's of equal length, divide the path up, as shown, using closely spaced concentric circles centred at the origin. Consider the pair Δ_j and Δ_k cut off by the same two circles. From the previous figure, we deduce that real components of $\tilde{\Delta}_j$ and $\tilde{\Delta}_k$ in the Riemann sum will cancel. But every Δ belongs to such a pair (or pairs), because if the closed loop L passes from the interior of a circle to the exterior, then in order to join up with itself back in the interior, it must recross the circle in the opposite direction somewhere else. **Thus the integral of $(1/z)$ along a closed loop is always imaginary.** The previous figure also implies that **this imaginary answer = i (net angle of rotation as the loop is traversed).** [b] The illustrated loop L does not enclose the origin, so the integral vanishes. [c] If the loop does wind around 0 once, then the integral is $2\pi i$.

the vertical and horizontal directions, as well as shrinking their lengths to $d\theta$ and (dr/r) , respectively.

Let us now see what happens if we stick all these $\tilde{\Delta}_j$ together for a closed loop such as L (see [8.13a]) that does not encircle the origin. In order to accomplish this we will forsake our previous choice of equal lengths for all the Δ_j , and instead divide the path up, as shown, using closely spaced concentric circles centred at the origin⁶. Consider the illustrated pair Δ_j , Δ_k lying between adjacent circles. That the Δ 's always *do* occur in such pairs is a consequence of L being a loop. For if

⁶ This only fails if part of L coincides with such a circle, but in that event we already know that the contribution to the integral is $i\Psi$.

L passes from the interior of a circle to the exterior, then in order to join up with itself back in the interior, it must recross the circle in the opposite direction somewhere else. Of course L may weave back and forth across a circle many times (e.g. at a, a^*, b, b^*, c, c^*), but the crucial point is that these crossings always occur in oppositely directed pairs.

In [8.13b] we see the consequence of this for R . From [8.12b] it's clear that for a pair such as $\tilde{\Delta}_j$ and $\tilde{\Delta}_k$, the horizontal components *cancel*. Since we have seen that *every* Δ belongs to such a pair, it follows that R will have no horizontal component for any closed loop, whether or not the origin is encircled. It also follows from [8.12b] that the height of this vertical Riemann sum is obtained by adding up all the signed angles that the Δ 's subtend. For a loop such as [8.13a], which does not encircle the origin, this sum is zero: as z traces out L its direction merely oscillates, rather than executing a complete revolution. Thus, as illustrated in [8.13b], R closes up on itself. On the other hand, if we translated L to any location where it encircled the origin then z would execute a complete revolution, and [8.13b] would change into [8.13c].

8.4.3 Winding Number

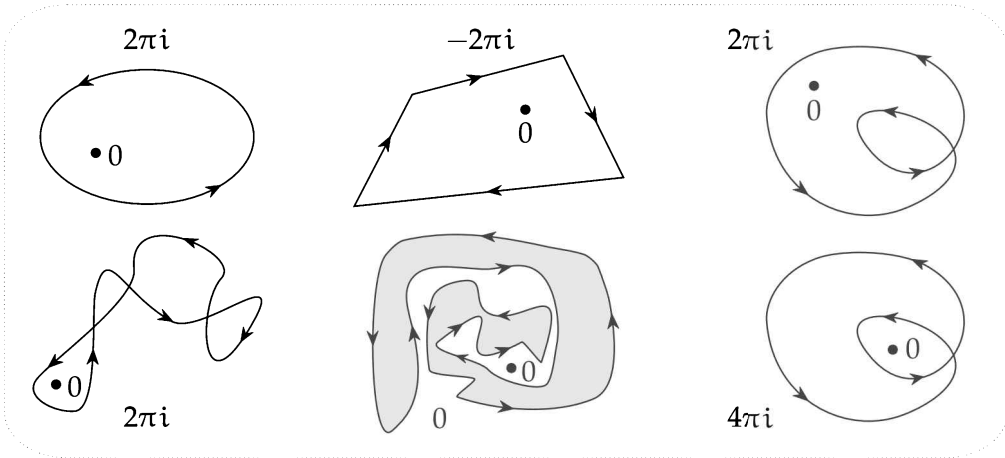
Let's recap. If a closed loop does not encircle the origin then the complex inversion mapping is analytic everywhere inside it, and in accord with Cauchy's Theorem the integral dutifully vanishes. If the origin *is* encircled, then the integral is no longer required to vanish by the theorem: the enclosed region now contains a point at which the mapping is not analytic. Indeed we found that for an origin-centred circle the answer was not zero, but $2\pi i$. Furthermore, the general investigation revealed that we would have obtained exactly the same answer if we had instead used an elliptical loop, or even a square loop. For if we distort the circle into one of these more general shapes, then all that happens to R is that it meanders about (illustrated in [8.13c]) on its net vertical journey to $2\pi i$, instead of marching straight there as it did in [8.11c].

We see that what really matters is not the shape of the loop, but rather its winding number about the origin. Thus we may summarize our findings tidily as follows: If L is *any* closed loop, then

$$\oint_L \frac{1}{z} dz = 2\pi i \nu(L, 0), \quad (8.6)$$

where the integral sign with a circle through it (which is a standard symbol) serves to remind us that we are integrating around a closed contour. Figure [8.14] shows various loops and the corresponding value of the integral of $(1/z)$ round each of them. Finally, note that (8.6) can easily be generalized [exercise] to

$$\oint_L \frac{1}{z - p} dz = 2\pi i \nu(L, p). \quad (8.7)$$



[8.14] The previous figure proves that $\oint_L (1/z) dz = 2\pi i \nu(L, 0)$.

8.5 Conjugation

8.5.1 Introduction

In the introduction we stressed that integration makes sense for any continuous complex mapping, regardless of whether or not it is analytic. However, the relatively lawless non-analytic functions give rise to integrals that behave less predictably than their analytic counterparts. In particular, Cauchy's Theorem has no jurisdiction here, and we therefore have no reason to anticipate path independence or, equivalently, vanishing loop integrals. As an example of this type of behaviour, we will show presently that the loop integral of the non-analytic conjugation mapping $z \mapsto \bar{z}$ yields the area enclosed by the loop. Assuming this result for the moment, let us use the examples $\bar{z}$ and $(1/z)$ to spell out more clearly the differences between the non-analytic and analytic cases.

In the analytic case, provided that the special point $z = 0$ was not enclosed, the loop integral vanished. Even when the integral of $(1/z)$ did *not* vanish, its possible values were still neatly *quantized* in units of $2\pi i$; one unit for each time the special point $z = 0$ was enclosed by the loop. As we will see later, this behaviour is typical, although a more general mapping may well possess *several* special points (at which analyticity breaks down) dotted about in the plane. Once again, the integral is not sensitive to the precise shape of the loop. Provided that none of the special points are enclosed by the loop, then the integral vanishes. However, if some of the points are enclosed, then each one makes its own distinctive contribution (generally not $2\pi i$) to the integral, one unit for each time it's encircled. The value of the integral is just the sum of these discrete contributions.

Contrast all this with our non-analytic example. The area of the loop (and hence the integral of $\bar{z}$) will almost never vanish. Furthermore, instead of being determined by stable topological properties, the value of the integral is sensitive to the detailed geometry of the loop. Finally, the value is not neatly quantized, but instead varies continuously as the loop changes shape.

8.5.2 Area Interpretation

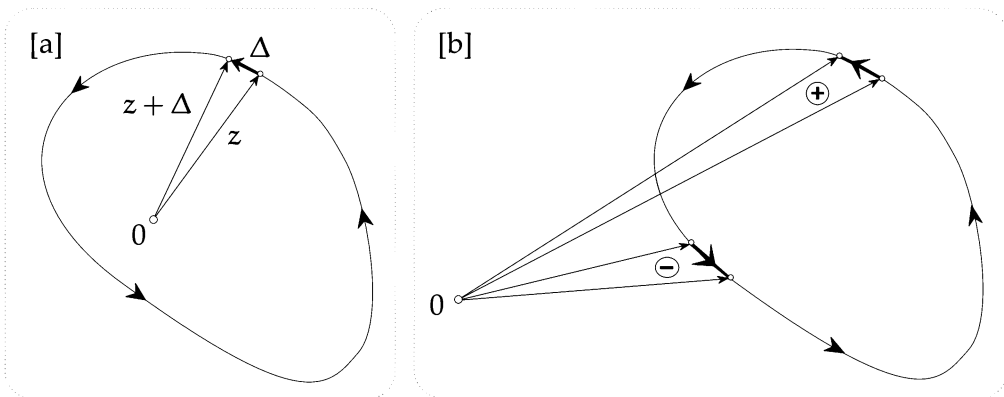
Let us now verify the area interpretation of the integral of $\bar{z}$. Recall from Chapter 1 that $\text{Im}(a\bar{b})$ is just twice the area of the triangle spanned by a and b . As z traces the loop L in [8.15a], think of the area it sweeps out as being decomposed into triangular elements, as illustrated. Thus

$$2 (\text{element of area}) = \text{Im}[(z + \Delta)\bar{z}] = \text{Im}[\bar{z}\Delta].$$

Adding these elements together, we obtain the imaginary part of the Riemann sum corresponding to the integral of $\bar{z}$. Thus we conclude that

$$\text{Im} \oint_L \bar{z} dz = 2 (\text{area enclosed}).$$

This result can be further simplified by noticing that $\bar{z}$ and $(1/z)$ both point in the same direction. It follows that we could draw a picture very similar to [8.12], the only difference being that to obtain $\tilde{\Delta}$ we would multiply by r instead of dividing by



[8.15] $\oint_L \bar{z} dz = 2i (\text{area enclosed by } L)$. [a] Using (1.20) on page 32, we see that $2 (\text{element of area}) = \text{Im}[(z + \Delta)\bar{z}] = \text{Im}[\bar{z}\Delta]$, so $\text{Im} \oint_L \bar{z} dz = 2 (\text{area})$. But $\bar{z}$ has the same direction as $(1/z)$, and the argument given in [8.12] easily generalizes, so **the integral is purely imaginary**, thereby proving the result. [b] The geometric meaning of the integral is unaltered if 0 lies outside L , for although the top triangle includes area that lies outside L , this extra area is cancelled by the negative area of triangles like the one at the bottom. Compare this to [1.22b] on page 33.

it. The argument that followed from [8.12] therefore remains valid, and we deduce that the integral of $\bar{z}$ around a closed loop is purely imaginary. Thus

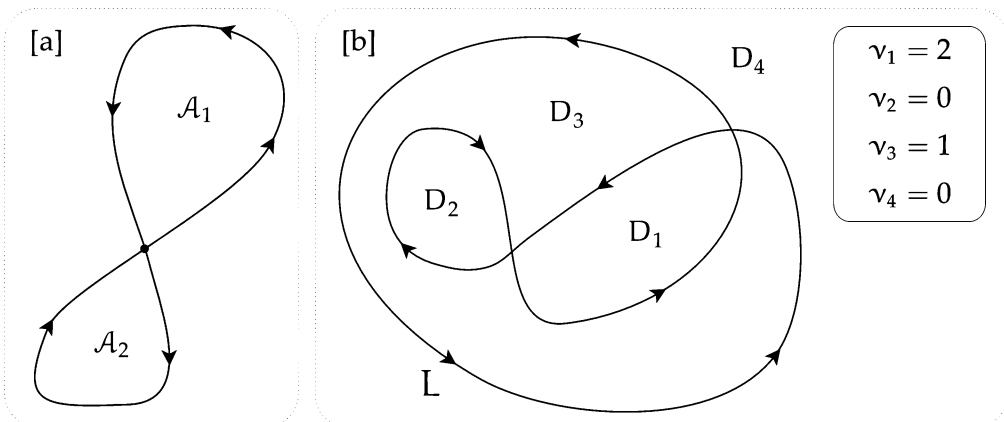
$$\oint_L \bar{z} dz = 2i (\text{area enclosed}). \quad (8.8)$$

Next we ask how this formula would change if the origin were *outside* the loop. Figure [8.15b] shows that the pleasing answer is, “Not at all!” The point is that the integral adds up the *signed* areas subtended by the Δ 's at the origin. On the far side, Δ carries z counterclockwise, yielding a positive element of area; but on the near side z is moving clockwise, yielding a negative element of area. When these are added, the unwanted area lying outside the contour simply cancels, leaving behind just the area enclosed.

As a simple example, consider a circle C of radius r centred at a , the equation of which is $r^2 = |z - a|^2 = (z - a)(\bar{z} - \bar{a})$. Solving this for $\bar{z}$, and using (8.7), we find that

$$\begin{aligned} \oint_C \bar{z} dz &= \bar{a} \oint_C dz + r^2 \oint_C \frac{1}{z - a} dz \\ &= 0 + r^2 2\pi i \\ &= 2i (\text{area enclosed}). \end{aligned}$$

From what we have done so far you might be inclined to think that the integral of $\bar{z}$ could never vanish for a nontrivial loop. That this is false can be seen from the figure eight loop in [8.16a]. This may be thought of as the union of two separate loops. The top one is traversed in a positive sense and correspondingly yields its ordinary area $\mathcal{A}_1$; but the bottom one is traversed in a negative sense and yields the



[8.16] [a] If L is a figure eight, then $\oint_L \bar{z} dz = 2i(\mathcal{A}_1 - \mathcal{A}_2)$. Note that if L were symmetrical, then the integral would vanish. **[b]** In general, $\oint_L \bar{z} dz = 2i \sum_{\text{inside}} v_j \mathcal{A}_j = 2i [2\mathcal{A}_1 + \mathcal{A}_3]$ in this example.

negative ($-A_2$) of its ordinary area. Thus the integral is $2i(A_1 - A_2)$, and if the loop were symmetrical then this would vanish.

8.5.3 General Loops

To finish off this example we will explain how the winding number concept can be used to evaluate the integral for more complicated loops. A typical loop such as [8.16b] will partition the plane into a number of sets D_j , and in the last chapter we defined the “inside” to consist of those D_j for which the corresponding winding number $v_j \neq 0$; the remaining D_j constitute the “outside”. We can now state the general result and leave you to ponder its truth:

$$\oint_L \bar{z} dz = 2i \sum_{\text{inside}} v_j A_j, \quad (8.9)$$

where A_j denotes the area of D_j . For example, in the case of [8.16b] we obtain $\oint_L \bar{z} dz = 2i [2A_1 + A_3]$. We hope you can already see why this example is correct, but a full explanation of the general formula (8.9) will be provided later in this chapter.

8.6 Power Functions

8.6.1 Integration along a Circular Arc

Having understood the integral of $(1/z)$ it is easy to understand the integrals of other powers. Once again let us begin by integrating along the circular arc K that was used in [8.11]. The result we will obtain is formally identical to the real result

$$\int_A^B x^m dx = \frac{1}{m+1} [B^{m+1} - A^{m+1}] \quad (m \neq -1),$$

but the difference is that in the complex case we can actually *see it!*

Figure [8.17a] shows a contour like that in [8.11a], while the transition from [8.11b] to [8.17b] represents the change from complex inversion to a general integer power $w = z^m$. Although the primary purpose of [8.17] is to convey the general argument, you will better understand its details if I tell you that it actually depicts the special case $m = 2$.

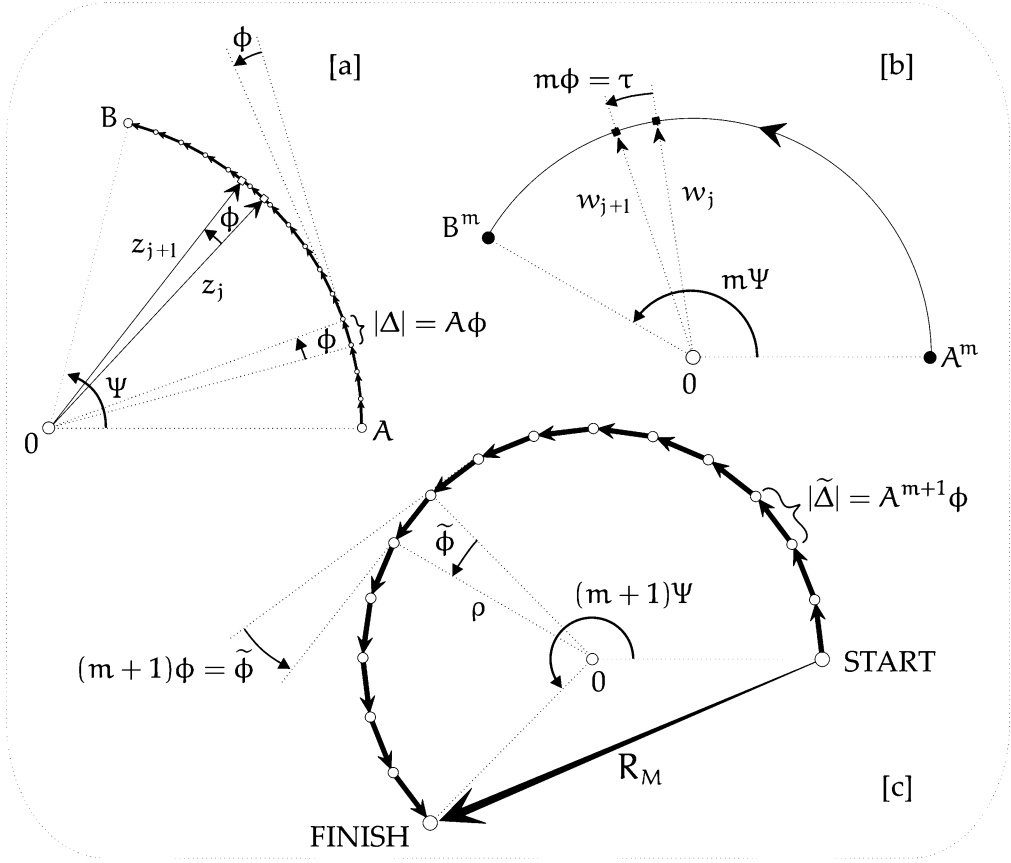
As z travels along K , w travels round an image circle of radius A^m , and with an angular speed that is m times as great. Thus

$$|\tilde{\Delta}| = A^m (A\phi) = A^{m+1}\phi,$$

and

$$\tilde{\phi} = \tau + \phi = m\phi + \phi = (m+1)\phi.$$

Since all the $\tilde{\Delta}$'s have the same length and the same turning angle, it follows that R_M is a polygonal approximation to an arc of a circle, the centre of which we have



[8.17] $\int_A^B z^m dz = \frac{1}{m+1} [B^{m+1} - A^{m+1}]$, because [a] $|\Delta| = A\phi$, [b] $\tau = m\phi$ and $|\tilde{\Delta}| = A^{m+1}\phi$, so [c] $\tilde{\phi} = \tau + \phi = (m+1)\phi$, and therefore $\rho = |\tilde{\Delta}|/\tilde{\phi} = A^{m+1}/(m+1)$.

chosen to place at the origin in [8.17c]. We will now determine the angle subtended by this arc, and also its radius.

The angle that each $\tilde{\Delta}$ subtends at the origin is the same as the turning angle $\tilde{\phi}$, namely, $(m+1)$ times the angle subtended by each Δ . Thus

$$\text{angle of FINISH} = (m+1)\Psi.$$

Also, if ρ is the radius, we see from the figure that

$$\rho \tilde{\phi} = |\tilde{\Delta}| \implies \rho = \frac{A^{m+1}}{m+1}.$$

We therefore conclude that if $m \neq -1$ then

$$\begin{aligned} R_M &= \text{FINISH} - \text{START} \\ &= \frac{1}{m+1} [A^{m+1} e^{i(m+1)\Psi} - A^{m+1}] \\ &= \frac{1}{m+1} [B^{m+1} - A^{m+1}] \end{aligned} \quad (8.10)$$

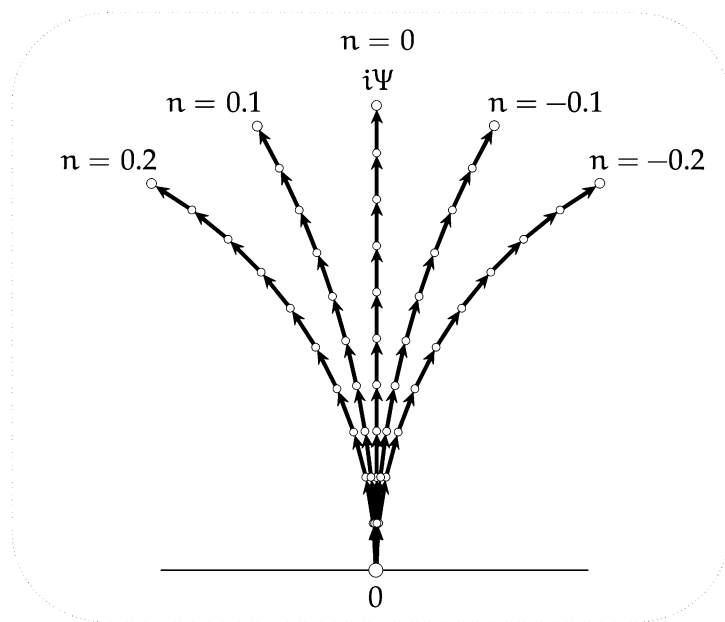
which, as promised, is formally identical to the real result. We hope you will agree that it's rather fascinating how we have been able to visualize this result in a way that would not have been possible in the real case.

As we have said, [8.17] actually depicts the concrete case $m = 2$, and before continuing you may care to sketch another case for yourself; $m = -2$ might be a fun one.

8.6.2 Complex Inversion as a Limiting Case*

As in ordinary calculus, we see that the case $m = -1$ (complex inversion) stands out from the crowd. Nevertheless, we can still understand the behaviour of this special power as a limiting case of other powers. With a little care about branches, the above result can be seen to persist even if we relax the requirement that m be an integer. As m gradually approaches -1 the radius ρ grows, and so R_M looks less and less curved; at the same time the lengths of the $\tilde{\Delta}$'s tend to ϕ . Thus in the limit that m tends to -1 we see that R_M will go straight up the imaginary axis to $i\Psi$. This is illustrated in [8.18]. The variable $n \equiv m + 1$ measures the difference between m and -1 , and it is therefore a good label for the Riemann sums shown in the figure.

Returning to the case of integer powers, we next observe that for a complete circular loop, there is a striking and fundamental difference between complex inversion and all other powers: if $m \neq -1$ then the integral *vanishes*. This is



[8.18] The integral of $(1/z)$ as a limiting case of the integral of z^m . Here, $n \equiv m - (-1)$ measures the “distance” of z^m from the ultimate target of z^{-1} . But there is also a fundamental difference between $(1/z)$ and all other powers: its integral around a complete circle is $2\pi i$, whereas the integral for all other powers is zero!

because R_M will now go round in a complete circle $|n|$ times [clockwise if $n < 0$; counterclockwise if $n > 0$], thereby returning to its beginning.

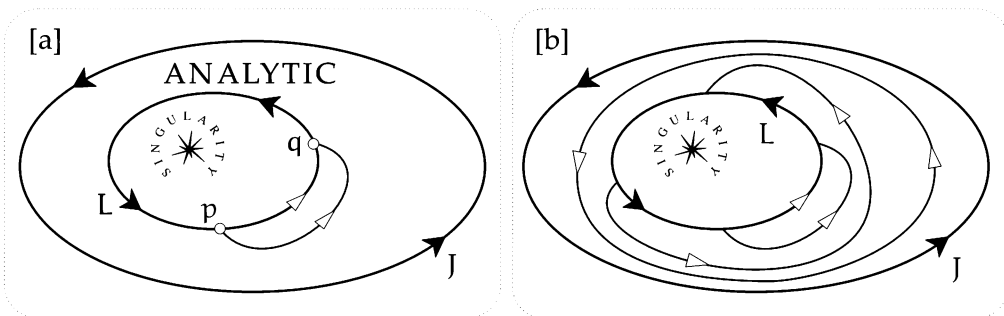
8.6.3 General Contours and the Deformation Theorem

Thus far we have only established (8.10) for the case where A and B are connected by a simple arc, but in fact it is true for almost any contour. Take the case $n > 0$ first. Since z^n is then analytic throughout the plane, it follows directly from Cauchy's Theorem that all contours will yield the same value. However, when $n < 0$ the situation is a little bit more subtle.

Just as complex inversion suffers a breakdown of analyticity at the origin, so too do all the other negative powers of z . Therefore Cauchy's Theorem only guarantees that two connecting paths yield the same integral *provided* that together they do not enclose the origin. For a loop that does enclose the origin, the integral is not required to vanish, and indeed in the case of z^{-1} it equals $2\pi i$.

Nevertheless, our direct evaluation reveals that for all other negative powers the integral round a circular loop *does* vanish, despite not being required to⁷. We will now derive a new form of Cauchy's Theorem that enables us to show that the vanishing of the integral is not a fluke resulting from the special circular shape of the loop.

Consider [8.19a]. The two loops J and L both encircle a singularity of some mapping, and so neither integral is required to vanish by Cauchy's Theorem. However, if the mapping is analytic in the shaded region lying between the loops, then we will now show that the two integrals must be equal. First consider the contribution



[8.19] Deformation Theorem. [a] Deform L by pushing the segment pq outward to form the illustrated bump. Since the mapping is analytic between the two paths connecting p and q , it follows from Cauchy's Theorem that both integrals are equal, and since the rest of L hasn't changed, it follows that **integral with bump = integral without bump**. [b] We can continue to deform L until it becomes J , and the value of the integral will not change.

⁷ In Chapter 11 we will give a physical explanation for this difference between complex inversion and the rest of the negative powers.

to the integral round L that comes from the piece between p and q . Suppose that we deform L slightly by replacing this segment by the bump in the figure. Since the mapping is analytic between the two paths connecting p and q , it follows from Cauchy's Theorem that both integrals are equal. Also, since the rest of L hasn't changed, it follows that *integral with bump* = *integral without bump*. All we need do now, to obtain the stated result, is to let the bump grow and change shape (see [8.19b]) until L has evolved into J .

The crucial idea is this

Deformation Theorem. *If a contour sweeps only through analytic points as it is deformed, the value of the integral does not change.* (8.11)

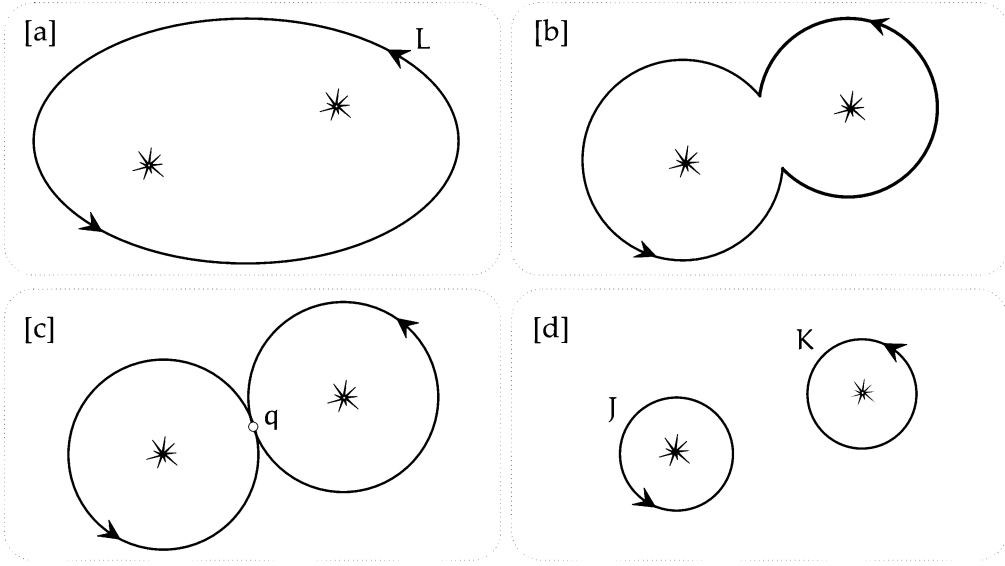
Thus, if you imagine the contour to be a rubber band, and the singularity to be a peg sticking out of the plane (thereby obstructing motion past it), the integral has the same value for all shapes into which the rubber band can be deformed.

We can immediately apply this Deformation Theorem to our problem. For if the mapping is a negative power of z other than z^{-1} then the established fact that the integral vanishes for a circular loop implies it continues to vanish for any loop into which the circle can be deformed without crossing the singularity at the origin. Thus formula (8.10) is path independent even for negative powers.

The Deformation Theorem also provides us with a much simpler derivation of the result (8.6) governing the general loop integral of the complex inversion mapping. Imagine taking a length of elastic string and winding it around an origin-centred circle ν times, finally joining the ends together to form a closed loop. From our earlier work, it follows that the value of the integral is then $2\pi i\nu$. But the Deformation Theorem says that this will be the value of the integral for any loop into which the elastic string may be deformed without being forced over the peg (singularity) at the origin. Finally, by the Hopf Degree Theorem, the loops into which it can so be deformed are those with winding number ν .

8.6.4 A Further Extension of the Theorem

Our 'dynamic' version of Cauchy's Theorem can be further extended to embrace mappings that have *several* singularities. Consider a loop L (see [8.20a]) encircling two singularities (pegs) of some mapping; the generalization to more singularities will be obvious. If we deform L without forcing it over a peg then we know that the integral will remain constant. The process [8.20a] $\rightarrow$ [8.20b] $\rightarrow$ [8.20c] is an example of such a deformation. The situation in [8.20c] is now rather interesting. The contour has become pinched together at q , and the value of the integral can be thought of as the sum of the two separate integrals taken round the touching circles. But now, by the same reasoning as usual, we may separately distort these circles so that [8.20c] $\rightarrow$ [8.20d]. Thus we conclude that



[8.20] Deformation Theorem with multiple singularities. We can deform L into $J + K$ without changing the value of the integral.

$$\oint_L f(z) \, dz = \oint_J f(z) \, dz + \oint_K f(z) \, dz. \quad (8.12)$$

To illustrate (8.12), consider $f(z) = 2/(z^2 + 1)$ which has singularities at $z = \pm i$. We can evaluate the integral round any loop C by noting this alternative expression:

$$f(z) = \frac{i}{z + i} - \frac{i}{z - i}.$$

Applying (8.7) therefore yields

$$\oint_C f(z) \, dz = 2\pi[\nu(C, i) - \nu(C, -i)].$$

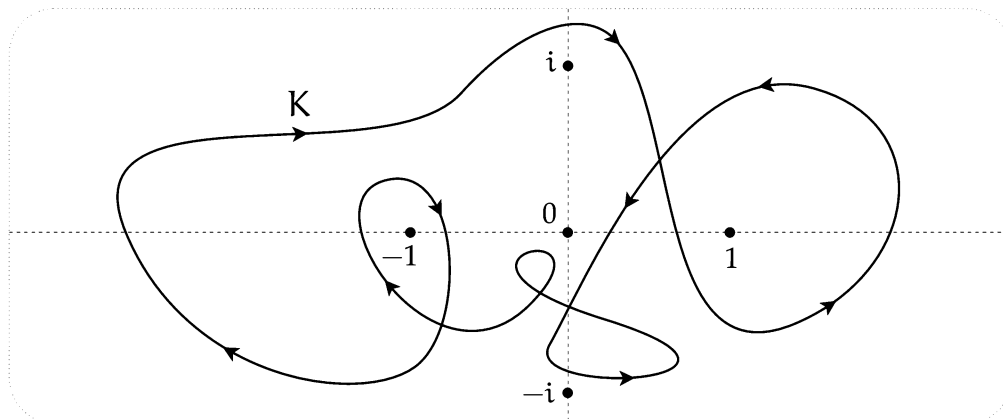
Assuming (as in [8.20a]) that L encloses both singularities, use this formula to verify (8.12) for this particular function.

8.6.5 Residues

Since we now possess a fairly complete understanding of the loop integrals of power functions, it is relatively easy to integrate simple rational functions: we need only find the decomposition into so-called *partial fractions*, and then integrate term by term. Indeed, this is precisely what we did in the example of the last paragraph.

Here is a slightly more complicated example: the integral of $f(z) = z^5/(z + 1)^2$ taken round the contour K in [8.21]. This has a second-order pole at $z = -1$, and by writing the numerator as $[(z + 1) - 1]^5$, we quickly find that

$$f(z) = -\frac{1}{(z + 1)^2} + 5 \left[\frac{1}{z + 1} \right] - 10 + 10(z + 1) - 5(z + 1)^2 + (z + 1)^3.$$



[8.21] The Residue of a singularity. Let $f(z) = z^5/(z+1)^2$, which has a second-order pole at $z = -1$. If $f(z)$ is decomposed into partial fractions then the only part that is not removed by integration is its “residue”—that which remains, defined to be the coefficient of the complex inversion term, $1/(z+1)$. This is the only part that does not integrate away into thin air. The value of the integral therefore has only two ingredients: the residue and the winding number, and the result is $2\pi i$ times their product, which turns out to be $-20\pi i$ for this particular function and contour.

But we know that the loop integral of powers other than -1 is zero, and so *only* the complex inversion term [in square brackets] can contribute. In detail,

$$\oint_K f(z) \, dz = 5 \cdot 2\pi i \, v(K, -1) = -20\pi i.$$

Thus the value of the integral has been determined by just two factors: the winding number of the loop, and the amount (i.e., coefficient) of complex inversion contained in the decomposition of the mapping. Because this latter number is the only part of the function that remains after we integrate, it is called the *residue* of the function at the singularity. Quite generally, the residue of $f(z)$ at a singularity s is denoted $\text{Res}[f(z), s]$. Thus in the above example, $\text{Res}[z^5/(z+1)^2, -1] = 5$.

In fact the residue concept has a significance that extends far beyond simple rational functions, as our next example will illustrate. We have previously [page 259] alluded to the remarkable fact that analytic functions are infinitely differentiable, or equivalently, that they can always be represented by a power series (Taylor’s) in the vicinity of a nonsingular point. For example, the Taylor series centred at the origin for $\sin z$ is

$$\sin z = z - \frac{1}{3!}z^3 + \frac{1}{5!}z^5 - \frac{1}{7!}z^7 + \cdots.$$

Clearly no such expansion can be possible at a singular point of a mapping. Nevertheless, we may recover an analogous result near singularities simply by broadening our notion of a power series to include *negative* powers. Such a series is called a *Laurent series*.

Consider $(\sin z)/z^6$. This is singular at the origin, but by simple division of the above Taylor series we obtain the following Laurent series in the vicinity of the singularity:

$$\frac{\sin z}{z^6} = \frac{1}{z^5} - \frac{1}{3!z^3} + \frac{1}{5!} \left[\frac{1}{z} \right] - \frac{1}{7!}z + \frac{1}{9!}z^3 - \dots$$

Once again, the residue of the function is defined to be the coefficient of the complex inversion term: $\text{Res}[(\sin z)/z^6, 0] = 1/5!$ in this case. If a power series converges at every point on a contour, then we may accept for the moment that it makes sense to integrate the series term by term. Once again we see that for a closed loop the sole contribution to the integral comes from the residue. For example, if K is the contour in [8.21] then

$$\oint_K \frac{\sin z}{z^6} dz = \frac{1}{5!} 2\pi i \nu(K, 0) = -\frac{2\pi i}{5!}.$$

The above examples of evaluating loop integrals in terms of residues are instances of *Cauchy's Residue Theorem*. We will return to these matters at the end of this chapter, and, in greater detail, in the following chapter. For the moment we simply remark that if a mapping possesses several singularities, then a residue can be attributed to each one.

8.7 The Exponential Mapping

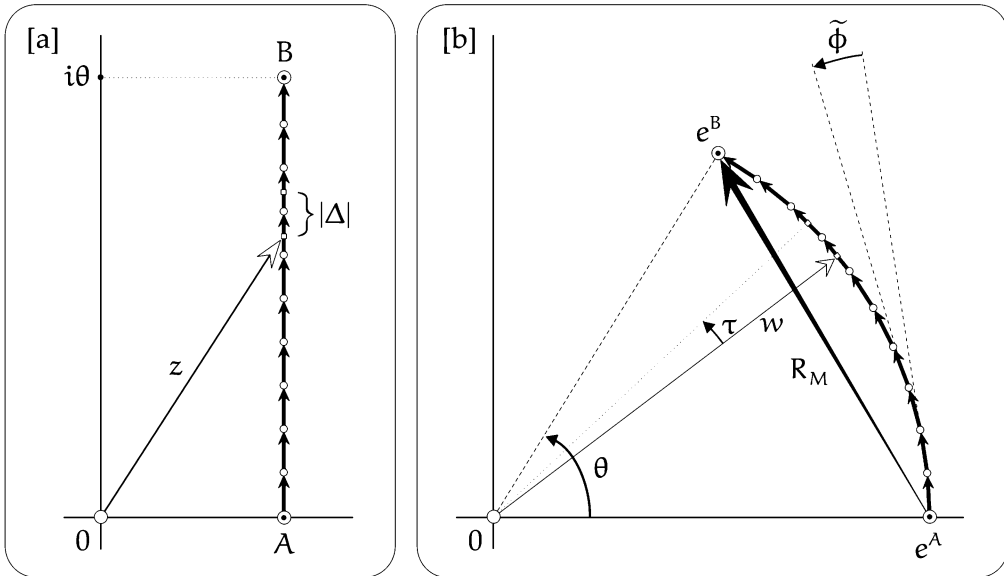
In the case of the exponential mapping the easiest contour along which to integrate is a vertical line-segment, say L (see [8.22a]). Once again we will find the result to be formally identical to its real counterpart:

$$\int_L e^z dz = e^B - e^A. \quad (8.13)$$

As z travels from A up to B in [8.22a], its image under $w = e^z$ will travel round the arc shown in [8.22b]. In order to verify (8.13), we will now show that (provided we choose to begin drawing R_M at e^A) *this arc is also the precise path taken by the Riemann sum*.

First note that since Δ_1 and w_1 are effectively vertical and horizontal, respectively, it follows that R_M will head off in the required vertical direction. Also observe that all the $\tilde{\Delta}$'s have the same length, namely, $|\tilde{\Delta}| = e^A |\Delta|$. Finally, since L has no bends (i.e., $\phi = 0$), $\tilde{\phi} = \tau = |\Delta|$. Because the $\tilde{\Delta}$'s all have the same length and turning angle, R_M will follow an arc of a circle. It only remains to show that if we begin drawing it at e^A , then this is the *same* arc as in [8.22b].

The next section will reveal the simplest way of seeing this, but the following direct argument is quite straightforward. We first verify that the two arcs have the



[8.22] Geometric evaluation of $\int_L e^z dz$. [a] Let L be the illustrated vertical segment from A to $B = A + i\theta$. [b] If we choose to begin drawing R_M at e^A then the illustrated arc traced by $w = e^z$ is also the precise path taken by the Riemann sum, and therefore $\int_L e^z dz = e^B - e^A$. First, all the $\tilde{\Delta}$'s have the same length, namely, $|\tilde{\Delta}| = e^A |\Delta|$. Since L has no bends (i.e., $\phi = 0$), $\tilde{\phi} = \tau = |\Delta|$. Because the $\tilde{\Delta}$'s all have the same length and turning angle, R_M will follow an arc of a circle, and it only remains to verify its radius. Try this yourself, or see the text for details.

same radius. The angle that each $\tilde{\Delta}$ subtends at the centre of its circle will be the same as its turning angle $\tilde{\phi}$. Therefore

$$\text{radius} = \frac{\text{arc}}{\text{angle}} = \frac{|\tilde{\Delta}|}{\tilde{\phi}} = e^A,$$

as required. Lastly, the total angle subtended by the arc at its centre is just the sum of all the $\tilde{\phi}_j = |\Delta_j|$, namely θ . The identity of the two arcs is thus established.

Since e^z is singularity-free, Cauchy's Theorem assures us that its loop integral always vanishes. Thus (8.13) must in fact be valid for *any* path from A to B .

8.8 The Fundamental Theorem

8.8.1 Introduction

Through specific geometric constructions, combined with the use of Cauchy's Theorem, we have already learnt a good deal about the integrals of some of the most important functions. However, there are two immediate problems still to be resolved: one is pragmatic, while the other is aesthetic.

The pragmatic one is that the formulae (8.10) and (8.13) are only known to hold for certain special configurations of the points A and B: the derivation of (8.10) assumes that they are equidistant from the origin; while for (8.13) they are assumed to be vertically separated. To be sure, our various forms of Cauchy's Theorem guarantee us that the integrals in question will continue to be path-independent, no matter what the locations of A and B. But the problem is that we haven't yet established that these path independent values will continue to be given by the same formulae as before. In this section we shall see that they are.

The aesthetic concern lies in the manner in which we derived path-independence for negative powers of z . Recall that we were only able to apply Cauchy's Theorem after having explicitly produced an example (a circle) of a loop integral that vanishes *in spite* of enclosing the singularity. Although this was neat enough in itself, one is left with the feeling that Cauchy's Theorem cannot be the most direct way of understanding a loop integral that continues to vanish in the presence of singularities.

A resolution of both these problems is provided by the so-called *Fundamental Theorem of Contour Integration*—a result that is formally identical to its similarly named counterpart in ordinary calculus. The naming of this theorem is not entirely appropriate, at least in the context of complex analysis. After all, so far we have managed quite well *without it*, suggesting that if this theorem is “Fundamental”, then Cauchy's must be “Super-Fundamental”!

8.8.2 An Example

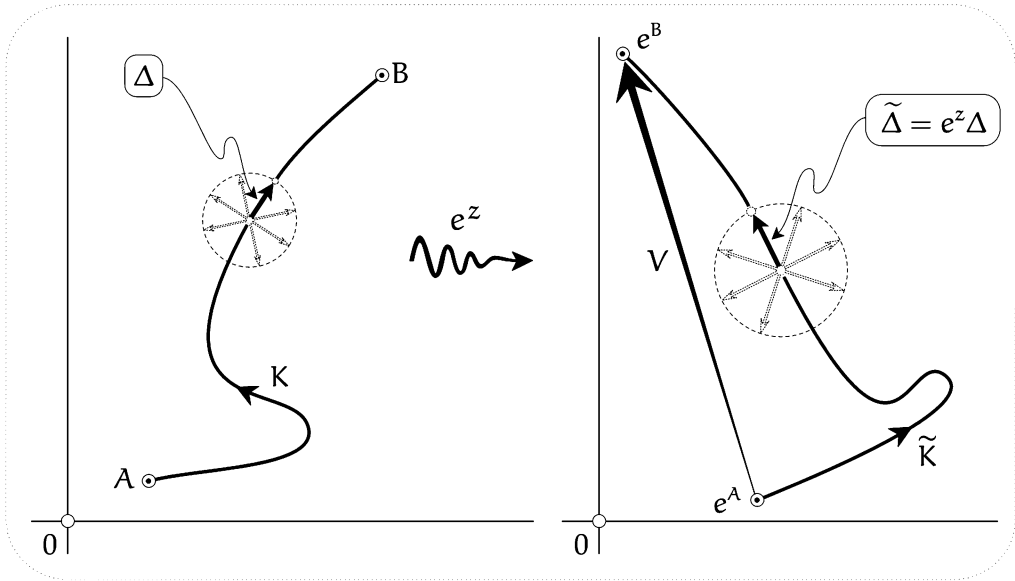
As our first example of this theorem, let us return to the exponential mapping of the last section in order to discover why (8.13) is valid for *any* pair of points, not just ones that are vertically separated. As so often happens in mathematics, all that is required is a very slight shift in viewpoint.

Figure [8.23] depicts a curve K (connecting a pair of typical points A and B) being mapped by e^z to the curve $\tilde{K}$ connecting e^A and e^B . Now let us forget (for a moment) all about integration and Riemann sums, and instead look at the figure from the point of view of *differentiation*.

All the little arrows emanating from a point on K will be mapped to images emanating from a point on $\tilde{K}$. In particular, if the arrow Δ is a little chord of K [tangent, in the limit that it shrinks], then its image $\tilde{\Delta}$ will likewise be a little directed chord of $\tilde{K}$. But for an *analytic* mapping, such as we are now considering, the original arrows are sent to their images by a simple amplitwist:

$$\tilde{\Delta} = (\text{amplitwist of } e^z) \cdot \Delta = e^z \Delta. \quad (8.14)$$

If we now add up all these vector chords of $\tilde{K}$ then we obtain the connecting vector V between its start and its finish. But (8.14) tells us that this vector V may also be



[8.23] The Fundamental Theorem in the case of e^z . Under $z \mapsto \tilde{z} = e^z$, the contour K connecting A to B maps to the curve $\tilde{K}$ connecting e^A to e^B , the connecting complex number from start to finish being $V = e^B - e^A$. But each Δ along K is amplitwisted to $\tilde{\Delta} = (\text{amplitwist of } e^z) \cdot \Delta = e^z \Delta$ along $\tilde{K}$, so $\tilde{K}$ simply is the Riemann sum, and therefore $\int_K e^z dz = V = e^B - e^A$.

interpreted as the Riemann sum corresponding to the integral of e^z along K . We have thus established the continued validity of (8.13) for all positions of A and B :

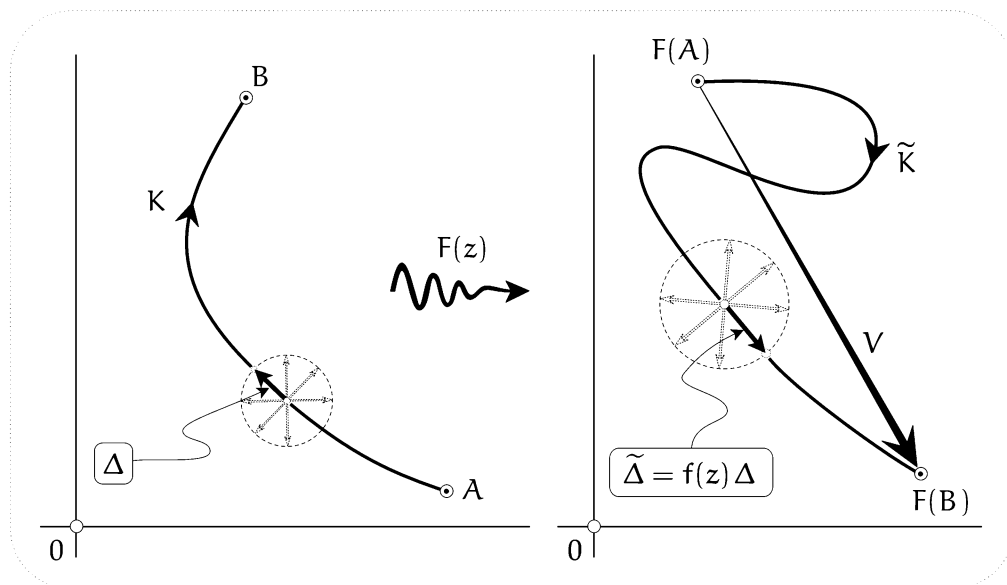
$$\int_K e^z dz = V = e^B - e^A.$$

To emphasize the path-independence of the construction, imagine choosing a different contour from A to B . The image curve (i.e., the new Riemann sum) will then take a different route from e^A to e^B , but of course the vector V will be quite unaffected.

8.8.3 The Fundamental Theorem

The Fundamental Theorem amounts to a restatement of the above idea in general terms. Suppose that we wish to evaluate $\int_K f(z) dz$ by the method above. We must seek an *analytic* mapping $F(z)$ whose amplitwist $F'(z)$ is given by $f(z)$. Assuming that such an F has been found [whether this animal even *exists* will be discussed shortly], we may then draw [8.24], which depicts the image curve $\tilde{K}$ under the mapping F . With the same terminology as before, (8.14) now becomes

$$\tilde{\Delta} = [\text{amplitwist of } F(z)] \cdot \Delta = f(z) \Delta.$$



[8.24] The Fundamental Theorem in the general case. If F maps K to $\tilde{K}$ and $F' = f$ then Δ is amplitwisted to $\tilde{\Delta} = [\text{amplitwist of } F(z)] \cdot \Delta = f(z) \Delta$, and therefore $\tilde{K}$ simply is the Riemann sum for the integral of f . Therefore, $\int_K f(z) dz = V = F(B) - F(A)$.

Just as before, we conclude that $\tilde{K}$ is actually the path taken by the Riemann sum of f , and that the vector V is once again the path-independent value of the integral:

$$\int_K f(z) dz = V = F(B) - F(A). \quad (8.15)$$

As in ordinary calculus, the function F cannot be unique, for $\hat{F} \equiv F + \text{const.}$ shares the same amplitwist. In terms of a figure like [8.24] this corresponds to the fact that the effect of $\hat{F}$ only differs from that of F by a translation; but this has no effect on the connecting arrow V .

In ordinary calculus, a real continuous function $f(x)$ *always* possesses an anti-derivative $F(x)$ for which $F' = f$. Of course it may not be easy to find, and it may not even be expressible in terms of elementary functions (e.g. $f(x) = e^{-x^2}$), but at least it exists. In the complex realm, on the other hand, we know that analytic functions are very special, and so we should not be surprised if the existence of such a function is no longer assured. Remember that when such an F exists, the integral of f is path-independent. It follows, for example, that no such function can exist for the non-analytic mapping $f(z) = \bar{z}$. Indeed, we may fall back on the still unproven result concerning infinite differentiability to see that, quite generally, analyticity of f is a necessary condition for the existence of F . For if F is analytic, then so too is its derivative F' , namely f .

When presented with the integral of a non-analytic function, it is therefore hopeless to seek an anti-derivative for use in (8.15)—no such function can exist. For the

special case $z \mapsto \bar{z}$ it is possible to extend the area interpretation (hence evaluation) to contours that are not closed, but for a general non-analytic mapping no such interpretation will be available. Although such integrals are of much less interest to us than those of analytic functions, in the next section we shall nevertheless find a method of evaluating them.

Before returning to more general considerations, let us give a couple more examples of the theorem in action. Consider $f(z) = z^2$. If we define $F(z) = \frac{1}{3}z^3$ then $F' = f$, and thus the path taken by the Riemann sum as we integrate along a contour will just be its image under $z \mapsto \frac{1}{3}z^3$. This allows us to look at the construction [8.17] in a new light. Recall that while this figure is concerned with the integration of a general power, it actually depicts the special case z^2 . In agreement with our new general result, we see that $z \mapsto \frac{1}{3}z^3$ does indeed map the contour in [8.17a] to its Riemann sum in [8.17c].

As an example of how the theorem also resolves our aesthetic concern over path-independence for negative powers of z , reconsider $f(z) = (1/z^2)$, for which we hope you did actually draw the analogue of [8.17], as suggested. Without appealing to Cauchy's Theorem (thus avoiding the attendant anxiety over the singularity at the origin), we see that since $(-1/z)' = (1/z^2)$, all contours⁸ between A and B will yield the same value for the integral:

$$\int_A^B \frac{1}{z^2} dz = \frac{1}{A} - \frac{1}{B}.$$

Notice, incidentally, that path-independence has allowed us to reinstate the familiar symbol $\int_A^B$ without fear of ambiguity.

Instead of having to use Cauchy's Theorem to extrapolate from the vanishing of the integral for a circle to its vanishing for more general loops, the conclusion is now immediate: since $B = A$ for a closed loop, the above expression vanishes.

8.8.4 The Integral as Antiderivative

We have seen that the existence of an *antiderivative* F (defined by $F' = f$) implies path-independence for the integral of f . We will now show, conversely, that path-independence implies the existence of F .

Let us first give another simple example of the Fundamental Theorem. Since $(\sin z)' = \cos z$, the integral of $\cos z$ will be path-independent, and if we integrate from the origin, for example, to a *variable* endpoint Z , then we obtain a well-defined function of Z :

$$F(Z) = \int_0^Z \cos z \, dz = \sin Z.$$

⁸ We exclude contours that actually pass *through* the singularity.

We note, without surprise, that this function is the antiderivative of $\cos Z$. If we began our integration at an arbitrary point, instead of at the origin, then the result would only differ by a constant, and so it would still be a perfectly good antiderivative.

In order to establish the claim of the first paragraph, it is only necessary for us to show that the above example is typical. If the integral of a mapping f is known to be path-independent, and A is an arbitrary fixed starting point, then we will show that

$$F(Z) \equiv \int_A^Z f(z) dz \quad (8.16)$$

is the antiderivative whose existence is sought. That is, we will verify that infinitesimal arrows emanating from a point P are merely amplitwisted to produce their images under this mapping F , and that the amplitwist at P is just $f(P)$ i.e., $F'(P) = f(P)$.

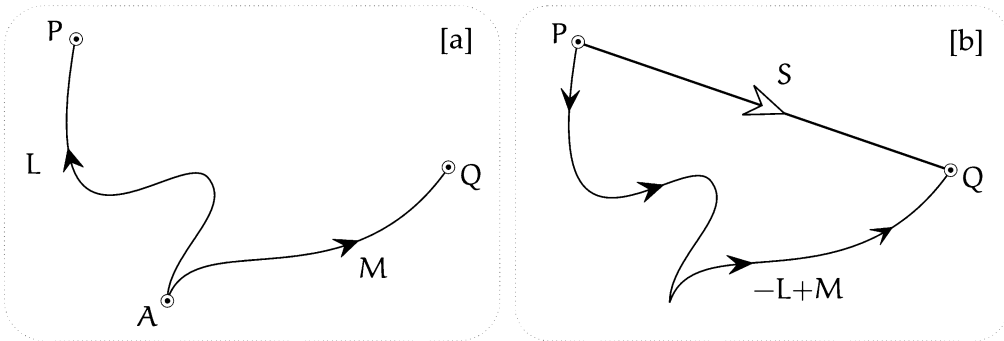
First we shall need a simple observation on differences of integrals. See [8.25a]. Two paths L and M are shown connecting the point A to the distinct points P and Q . We know that for any function $f(z)$,

$$\int_M f(z) dz - \int_L f(z) dz = \int_{-L+M} f(z) dz.$$

The path $(-L + M)$ for the right-hand integral is the round-about route from P to Q shown in [8.25b]. But if the integral is known to be path-independent then we may replace this path with the straight one S . Returning to the notation of (8.16), we thus have

$$F(Q) - F(P) = \int_S f(z) dz.$$

In the limit that Q coalesces with P , S becomes (with a minor abuse of terminology) an infinitesimal 'vector' Δ emanating from P , and its image $\tilde{\Delta}$ under F will be given by the left-hand side of the above equation. Thus



[8.25] [a] Given a fixed point A , suppose we evaluate $\int_M f(z) dz$ and $\int_L f(z) dz$. **[b]** Then $\int_M f(z) dz - \int_L f(z) dz = \int_{-L+M} f(z) dz$ is an integral along an indirect route from P to Q . If the integral is path-independent, we can replace the indirect route with the direct route S . Defining $F(Z) \equiv \int_A^Z f(z) dz$, we deduce that $F(Q) - F(P) = \int_S f(z) dz$.

$$F : \Delta \mapsto \tilde{\Delta} = \int_{\Delta} f(z) dz.$$

But if Δ is infinitesimal then the above integral equals $f(P) \Delta$, thereby establishing the original claim:

$$F : \Delta \mapsto \tilde{\Delta} = f(P) \Delta.$$

We may now appeal to Cauchy's Theorem to relate the required path-independence to analyticity. If f is analytic throughout some region then its integral is sure to be path-independent, and therefore F will exist. In other words, *every analytic mapping must itself be the derivative of another analytic mapping*.

We conclude this subsection with an interesting application of the analyticity of F . In the previous subsection we completely ignored our previous geometric constructions, and instead appealed to the Fundamental Theorem to show, for example, that

$$\int_A^B z^2 dz = \frac{1}{3}(B^3 - A^3),$$

for *all* positions of A and B . This was apparently a clear improvement on [8.17] where such formulae were merely established in the special case that A and B were equidistant from the origin. However, we will now see that analyticity makes it possible, paradoxically, for this special case to contain the general case.

Consider these two functions:

$$F(Z) = \int_A^Z z^2 dz \quad G(Z) = \frac{1}{3}(Z^3 - A^3),$$

both of which we now recognize as being analytic. From [8.17] we know that if Z moves along the origin-centred circle passing through A then

$$F(Z) = G(Z).$$

But by the uniqueness property of analytic functions [page 284] this identity must continue to hold even if Z wanders *off* the circle, thereby establishing the general result.

By applying exactly similar reasoning to the exponential mapping, we may likewise extrapolate the validity of (8.13) for vertically separated points (established by [8.22]) to deduce that

$$\int_A^Z e^z dz = e^Z - e^A,$$

even if Z wanders off the vertical line through A .

8.8.5 Logarithm as Integral

In the light of the Fundamental Theorem, we are inclined to jump to the conclusion that because $(\log z)' = (1/z)$,

$$\int_1^Z \frac{1}{z} dz = \log Z, \quad (8.17)$$

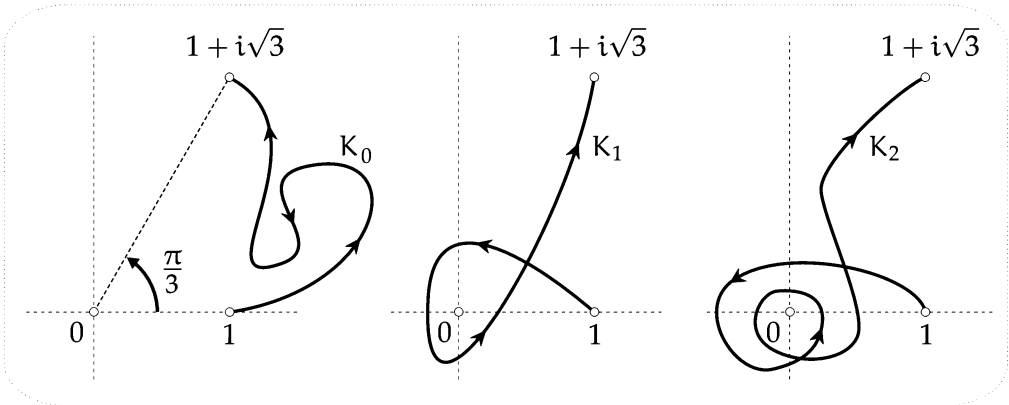
just as in real analysis. In a sense, this *is* correct, but a little care is required.

The subtlety is, of course, that the singularity at the origin causes the integral of $(1/z)$ not to be single-valued. Thus we must specify the contour K from 1 to Z before the integral in (8.17) becomes well defined. On the other hand, until we choose one of the infinitely many values $\theta(Z)$ for the angle of Z , the RHS of (8.17) is *also* not well defined. These two difficulties now cancel each other out in the following way.

In [8.26] we have drawn three different contours for the specific case $Z = 1 + i\sqrt{3}$. If we let $\theta_K(Z)$ stand for the net rotation as we follow K , then

$$\begin{aligned} \theta_{K_0}(Z) &= (\pi/3) \\ \theta_{K_1}(Z) &= (\pi/3) + 2\pi \\ \theta_{K_2}(Z) &= (\pi/3) + 4\pi. \end{aligned}$$

In a sense, including the contour in the definition of angle has rendered it single-valued. Notice that this definition does not depend on the precise shape of K , but only on how many times the origin is encircled.



[8.26] Why $\int_1^Z (1/z) dz = \log Z$ is a multivalued function. From [8.12] we deduce that $\int_1^Z (1/z) dz = \log_K(Z) = \ln|Z| + i\theta_K(Z)$, where $\theta_K(Z)$ denotes the net change in the angle as z traverses K . So the infinitely many values of $\log Z$ arise from the number of times K loops around 0 before finally arriving at Z , each extra revolution adding $2\pi i$ to the value of $\log Z$.

In this way, we may absorb the means of reaching Z into the definition of $\log(Z)$ in order to obtain a single-valued answer:

$$\log_K(Z) = \ln|Z| + i\theta_K(Z).$$

The unambiguously correct version of (8.17) then reads,

$$\int_K \frac{1}{z} dz = \log_K(Z).$$

Of course, the multiple-valued nature of $\log$ has merely been disguised, not done away with. Nevertheless, by pursuing the above idea one is led to consider so-called *Riemann surfaces*, whereby multifunctions *can* be rendered single-valued. But that is a story for another day.

8.9 Parametric Evaluation

When more elegant means are not available it is nevertheless possible (in principle) to evaluate a contour integral by expressing it in terms of ordinary real integrals. We shall now briefly describe and illustrate this method.

The basic idea is to think of the contour L as being traced by a moving particle whose position at time t is $z(t)$. Next, instead of building the Riemann sum (hence the integral) from very small vectors that are chords of L , we may equally well use very small vectors that are *tangent* to L . This is done using the tangential complex velocity $v = \frac{dz}{dt}$: the chord representing the movement during the instant of time δt may be replaced by the tangential vector $v \delta t$. Thus if L is traced out during the time interval $a \leq t \leq b$, then

$$\int_L f[z] dz = \int_a^b f[z(t)] v dt.$$

For example, suppose that L is one counterclockwise circuit of the circle with radius ρ and centre q , and that $f[z] = \bar{z}$. We know from our earlier work that the answer should be $2\pi i \rho^2$. Since $z(t) = q + \rho e^{it}$ ($0 \leq t \leq 2\pi$) and $v = i\rho e^{it}$, we obtain

$$\begin{aligned} \int_L \bar{z} dz &= \int_0^{2\pi} (\bar{q} + \rho e^{-it}) i\rho e^{it} dt \\ &= i\rho\bar{q} \int_0^{2\pi} (\cos t + i \sin t) dt + i\rho^2 \int_0^{2\pi} dt \\ &= 2\pi i \rho^2, \end{aligned}$$

as anticipated.

Naturally, the point of this method is not to confirm previously known results, but rather to evaluate integrals that we couldn't do before. For example, with the

same contour, but with $f[z] = \bar{z}^2$, the answer can no longer be guessed. However, you should now find it easy to discover that the answer is $4\pi i \bar{\rho}^2$.

By way of contrast with the non-analytic examples above, and as further practice with this method, confirm (using the same contour L) that $\int_L z^2 dz = 0$, as predicted by either Cauchy's Theorem or the Fundamental Theorem. Likewise, confirm that $\int_E z dz = 0$, where E is an origin-centred ellipse. [*Hint*: recall that $z(t) = p e^{it} + q e^{-it}$ moves on such an ellipse.]

For our last examples, take the contour to be a section of the parabola $y = x^2$ between 0 and $1 + i$; in temporal terms this can be represented as $z(t) = t + it^2$ ($0 \leq t \leq 1$). Integrate z along this contour, first using the Fundamental Theorem, then parametrically. Likewise, use the Fundamental Theorem to evaluate the integral for e^z . By equating the imaginary part of your answer with the imaginary part of the parametric evaluation, deduce that

$$\int_0^1 (2t \cos t^2 + \sin t^2) e^t dt = e \sin 1.$$

This result can be verified easily [exercise] without using complex numbers. Later, though, we shall meet real integrals that cannot readily be evaluated by such ordinary means, but which suddenly do become easy when viewed as arising from a *complex* integral.

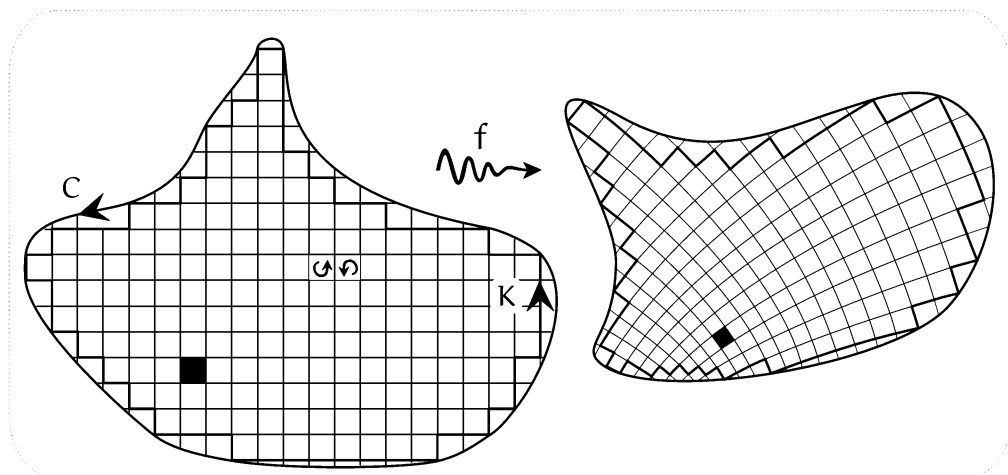
8.10 Cauchy's Theorem

8.10.1 Some Preliminaries

Having repeatedly witnessed the utility of Cauchy's Theorem in this chapter, it is perhaps time that we checked that it is *true*! We begin with the case where the contour C is a "simple" closed curve, i.e., without self-intersections. See [8.27].

We have filled the interior of C with a grid of small squares, of side length ϵ , aligned with the real and imaginary axes. We have then shaded all those squares that lie wholly within C , and taken the contour K to be the boundary of this shaded region, traversed counterclockwise. Because we have drawn relatively large squares (in order to make the picture clear), K is presently only a crude approximation to C . However, as we let ϵ shrink, the shaded region fills the interior of C ever more completely, and K follows C ever more precisely. Thus, in order to see whether or not the integral of a mapping f along C vanishes, it is sufficient to instead investigate the behaviour of the integral of f along K , as ϵ shrinks to zero. [This is justified in greater detail in Ex. 20.]

Next we seek to relate this integral along K to the behaviour of the mapping *inside* the shaded region that it bounds. Consider the sum of all the integrals of f taken counterclockwise round each of the infinitesimal shaded squares. This



[8.27] Cauchy's Theorem (step one): Fill the interior of C with a grid of small (ultimately vanishing) squares, of side length ϵ , aligned with the real and imaginary axes. The sum of the integrals around all these squares yields the integral around the boundary, K , for every interior edge is matched with an oppositely directed abutting edge that cancels it.

counterclockwise sense of integration is illustrated in [8.27] for two adjacent squares. When we add the integrals from these two squares, their common edge is traversed *twice*, once in each direction, and hence the integrals along it *cancel*. But this is true of every edge that lies in the shaded region, so that when we sum the integrals for all the shaded squares, the only edges that do not self-destruct in this manner are those that make up K :

$$\oint_K f(z) dz = \sum_{\text{shaded squares}} \oint_{\square} f(z) dz. \quad (8.18)$$

The investigation of the integral of f along C has thus been reduced to the study of the local effect of f on infinitesimal squares in the interior region.

It should be stressed that the discussion thus far is equally applicable to non-analytic and analytic mappings. For example, with $f(z) = \bar{z}$, (8.18) simply says [see (8.8)] that the area inside K is the sum of the areas of the shaded squares. In order to understand Cauchy's Theorem, we must specialize to the case (illustrated in [8.27]) where the local effect of f is an amplitwist throughout the interior of C . First, though, let us try to guess how the magnitude of a typical integral in the above summation will depend on ϵ (as the squares shrink) for a *general* mapping.

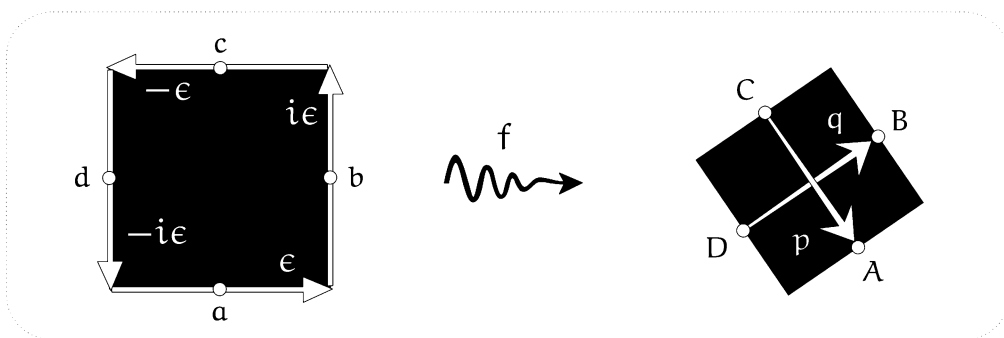
Experience with real integration, as well as the inequality (8.5), might lead one to guess that the integral round an infinitesimal square would die away at the same rate as its perimeter, that is, as ϵ . This is false. The fact that the square is a *closed* contour, together with the fact that complex integration is a type of *vectorial* summation, implies that the integral must decay much faster than this. In the above example of conjugation, we know that the exact value of each term is $2i\epsilon^2$, and this

leads us to the correct guess, namely, that the terms die away as the *square* of ϵ . We shall verify this in detail shortly, but for the moment the following rough argument will suffice.

We know that for a general mapping, the integral round K —hence the summation in (8.18)—will be nonzero and finite. This leads us to believe that each term must die away with the reciprocal dependence on ϵ as governs the growth of the number of terms in the series. But the number of terms grows as (*fixed area inside C , divided by the area of each square*), that is as $(1/\epsilon^2)$. Thus the magnitude of each term is expected to die away as ϵ^2 . If our original guess had been correct, the order of the sum in (8.18) would have been $\epsilon (1/\epsilon^2)$, yielding an infinite result as the squares shrunk. Conversely, any contributions to the terms involving powers of ϵ *greater* than two, cannot have any influence on the final result.

8.10.2 The Explanation

Let us return to [8.27] and to the explanation of Cauchy's Theorem. The analytic mapping f amplitwists the infinitesimal shaded squares on the left to the infinitesimal squares on the right, and [8.28] shows a magnified view of a typical such square and its image (the black ones in [8.27]). According to our especially accurate midpoint Riemann sum (R_M), the integral along the bottom edge of this square can be approximated by the single term $A \epsilon$: the image of the midpoint a , times the number along this edge. This conforms to our first, wrong guess concerning the dependence on ϵ of the complete integral round the square. But if we now add this to the integral along the *opposite* edge, the answer is



[8.28] Cauchy's Theorem explained. According to our especially accurate midpoint Riemann sum, R_M , the integral along the bottom edge of this square can be approximated by the single term $A \epsilon$: the image of the midpoint a , times the number along this edge. If we add this to the integral along the opposite edge, the answer is $A \epsilon + C (-\epsilon) = (A - C) \epsilon = p \epsilon$. Likewise, the contribution from the remaining two edges is also of order ϵ^2 , namely, $(B - D) i \epsilon = i q \epsilon$. But since f is locally an amplitwist, the image is a square, and so $i q = (q \text{ rotated through a right angle}) = -p$. Therefore, $\oint_{\square} f(z) dz = \epsilon (p + i q) = 0$.

$$A \epsilon + C(-\epsilon) = (A - C) \epsilon = p \epsilon.$$

Even if f is merely differentiable in the real sense, rather than locally an amplitwist, $|p|$ will still be proportional to ϵ , and the magnitude of $p\epsilon$ will therefore be proportional to ϵ^2 , as anticipated. Likewise, the contribution from the remaining two edges is also of order ϵ^2 , namely, $(B - D) i\epsilon = iq \epsilon$.

Perhaps you have already seen the light: if f is locally an amplitwist, the image is a *square*, and so

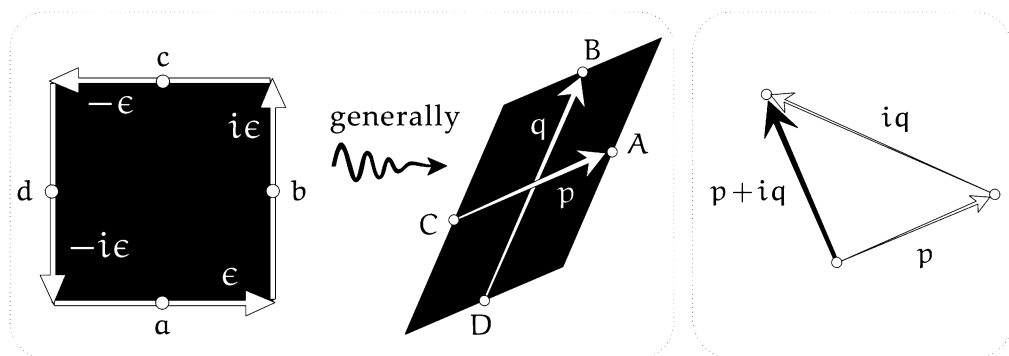
$$iq = q \text{ rotated through a right angle} = -p$$

$$\Rightarrow \oint_{\square} f(z) dz = \epsilon(p + iq) = 0. \quad (8.19)$$

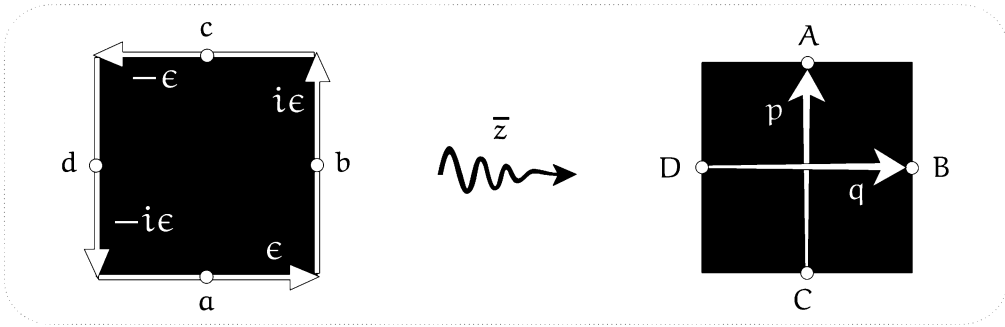
We conclude from (8.18) that the vanishing of loop integrals for analytic mappings is indeed the nonlocal manifestation of their local amplitwist property!

Contrast this with non-analytic mappings. See [8.29]. Provided that a mapping is differentiable in the real sense, we know [see page 236] that its local effect is expansion (by different factors) in two perpendicular directions, followed by a twist. Thus the image of an infinitesimal square will generally be a parallelogram; p and q will not have equal length, nor will they be orthogonal. As we see, p and iq no longer cancel, and $\epsilon(p + iq)$ is of order ϵ^2 . When we add up the terms of (8.18), of order $(1/\epsilon^2)$ in number, the answer will therefore be nonzero and finite.

Conjugation provides a particularly striking example of this noncancellation for non-analytic mappings. See [8.30]. In the terms of the previous paragraph we could say that its expansion factors are everywhere 1 and -1 (in the horizontal and vertical directions), and that its twist is zero. The image of the square is *again* a square, but there is a crucial difference between [8.28] and [8.30]. Because this mapping is *anticonformal*, it reverses the orientation of the square, and we see that



[8.29] The non-analytic case. The shrinking square is ultimately mapped to a (non-square) parallelogram. As we see, p and iq no longer cancel, and $\epsilon(p + iq)$ is of order ϵ^2 . When we add up the terms of (8.18), of order $(1/\epsilon^2)$ in number, the answer will therefore be nonzero and finite.



[8.30] $\oint_{\square} \bar{z} dz = 2i(\text{area of } \square)$, because $\epsilon(p + iq) = \epsilon(i\epsilon + i\epsilon) = 2i\epsilon^2$.

$$\oint_{\square} \bar{z} dz = \epsilon(p + iq) = \epsilon(i\epsilon + i\epsilon) = 2i\epsilon^2 = 2i(\text{area of } \square).$$

Returning to questions of analyticity, and comparing (8.19) with [8.29], we obtain a converse to Cauchy's Theorem. If all the loop integrals of f are known to vanish, then, in particular, they will vanish for infinitesimal squares, such as the one on the left of [8.29]. Thus, $p + iq = 0$. But it is clear that this can *only* happen if the image is another infinitesimal square with the same orientation as the original (cf. [8.30]). Thus the local effect of f must be an amplitwist. This converse is called *Morera's Theorem*.

As with other new ideas in this book, we have not attempted to present the arguments in rigorous form; "insight", not "proof", is ever our watchword. For example, consider these objections (ascending in severity) to the geometric argument of [8.27] and [8.28]: no matter how small the square, the sides of the image will not be *perfectly* straight (though they will meet in perfect right angles); the midpoint a will not be mapped to the *exact* midpoint of the image; and despite the undoubted accuracy of R_M for a very small contour, it will not yield the *exact* value of the integral.

Nevertheless, it seems plausible that [8.28] and its associated reasoning remain unimpeached when it comes to the evaluation of the *dominant* ϵ^2 contribution. Indeed, the example of [8.30] lends at least some credence to the irrelevance of the above objections, for in that case we know the answer is correct to this order of ϵ . [In fact it comes out exactly right, but that is a fluke.] More generally, recall that parametric evaluation revealed that the real and imaginary parts of any contour integral can be expressed as ordinary real integrals. This means that we may carry over to the complex realm our previous determination (8.3) of the error induced by R_M in real analysis. Thus each of the integrals along the four edges of the square will differ from their R_M -values by an amount that dies away at least as fast as ϵ cubed [cf. Ex. 21 and Ex. 22]. But as we have previously argued, as ϵ shrinks to zero, such contributions can have no effect on the sum in (8.18). Although we shall not dwell on them, other objections can be treated in a similar way.

8.11 The General Cauchy Theorem

8.11.1 The Result

Consider a mapping f that is analytic except at the singularity marked in [8.31]. Must the integral of f round K vanish, or not? You see the problem. For a simple loop without self-intersections (such as in [8.27]) it is perfectly clear whether a singularity is lurking inside, and consequently whether Cauchy's Theorem applies. But in [8.31], it is not even clear what "inside" means, let alone how this might relate to Cauchy's Theorem.

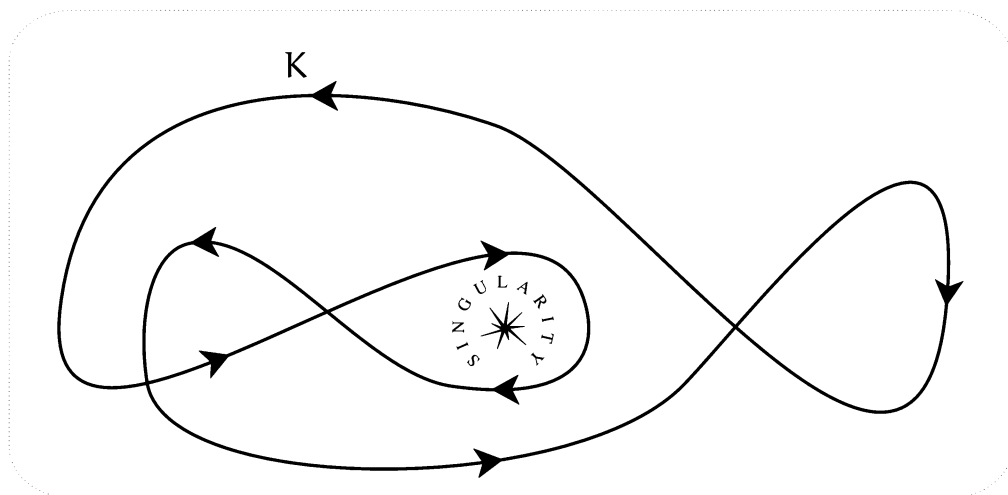
Recall that we encountered such problems before when trying to integrate $\bar{z}$ round complicated loops [see (8.9), as well as the discussion on p. 387]. Our solution was to define the "inside" to be all the points for which the winding number does not vanish, and conversely, the "outside" to be all the points for which it does vanish.

With these definitions in place, the completely general version of Cauchy's Theorem is stunning in its simplicity:

If an analytic mapping has no singularities "inside" a loop, its integral round the loop vanishes. (8.20)

This section is devoted to understanding this beautiful result.

First let us answer our opening question. In [8.31], K does not wind around the singularity, and therefore (according to the theorem) the integral should vanish. In



[8.31] As in Section 7.1.2, we define the "inside" of a closed loop to be the set of points for which the winding number is not zero. Then if an analytic mapping has no singularities inside a loop, its integral round the loop vanishes, as it does in this example.

the process of understanding this particular instance of the theorem we shall be led to a completely general argument for its validity.

8.11.2 The Explanation

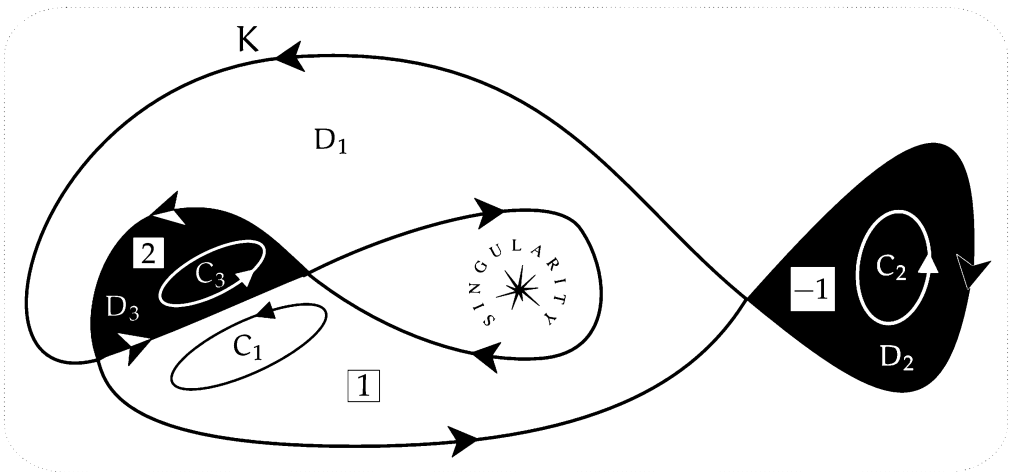
As in [8.16b], the contour K in [8.31] partitions the plane into a number of disjoint regions; in particular, the inside of K is made up of D_1 , D_2 , and D_3 . See [8.32]. Let C_j be the boundary of D_j , traversed counterclockwise. So as not to clutter up the picture, instead of actually drawing these contours in [8.32], we have merely indicated (with ellipses) their common counterclockwise sense. Also shown (in boxes) are the winding numbers of K around each of the regions D_j . Since there are no singularities inside the D_j that make up the inside of K , our basic version of Cauchy's Theorem applies to each of the simple contours C_j , and we have

$$\oint_{C_j} f(z) dz = 0. \quad (8.21)$$

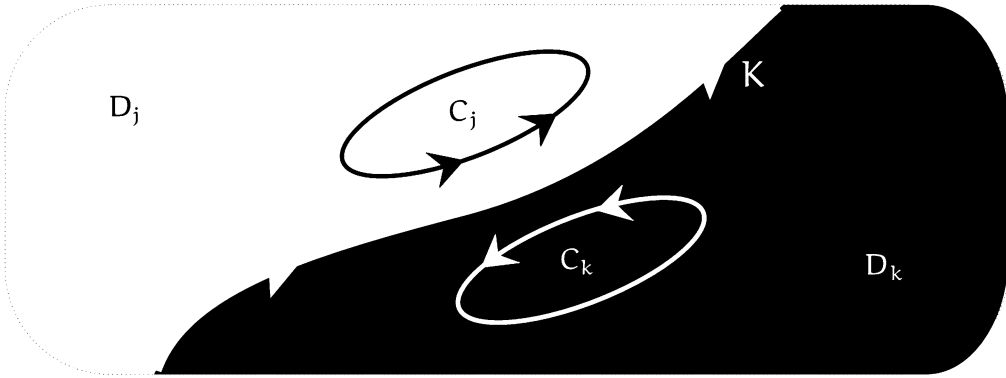
Now comes the crucial observation. The integral round K can be expressed as a linear combination of the integrals round the C_j 's that bound the interior D_j 's. In the case of [8.32],

$$\oint_K f(z) dz = \oint_{C_1} f(z) dz - \oint_{C_2} f(z) dz + 2 \oint_{C_3} f(z) dz. \quad (8.22)$$

Consider, for example, the contour C_3 , for which the counterclockwise sense happens to agree with the direction of K . On the other hand, C_1 traverses this portion of K in the *opposite* direction. Consequently, in (8.22), we end up integrating twice in the correct direction, and once in the opposite direction; the net result is to integrate



[8.32] $K = \sum_j v_j C_j = C_1 - C_2 + 2 C_3$ in this example. [The explanation is in the next figure.]



[8.33] Proof that $K = \sum_j \nu_j C_j$. Using the “crossing rule” (7.1) on page 388, we see that $\nu_j = \nu_k + 1$. Thus the contour C_j in the direction of K will always occur precisely one more time than the contour C_k in the opposite direction—the net result is that K is traversed once in the correct direction.

along this part of K once in the correct direction. You should check for yourself that all of K is correctly accounted for in this way. Substituting (8.21) into (8.22), we have confirmed the prediction of the general theorem for this particular contour.

Since (8.22) is clearly true for *any* function f , we may abstract it away and write the equation as

$$K = C_1 - C_2 + 2 C_3 .$$

Notice that the coefficient of C_j in this sum is none other than the *winding number* ν_j of K about D_j , and that we may therefore rewrite the previous equation as

$$K = \sum_j \nu_j C_j . \quad (8.23)$$

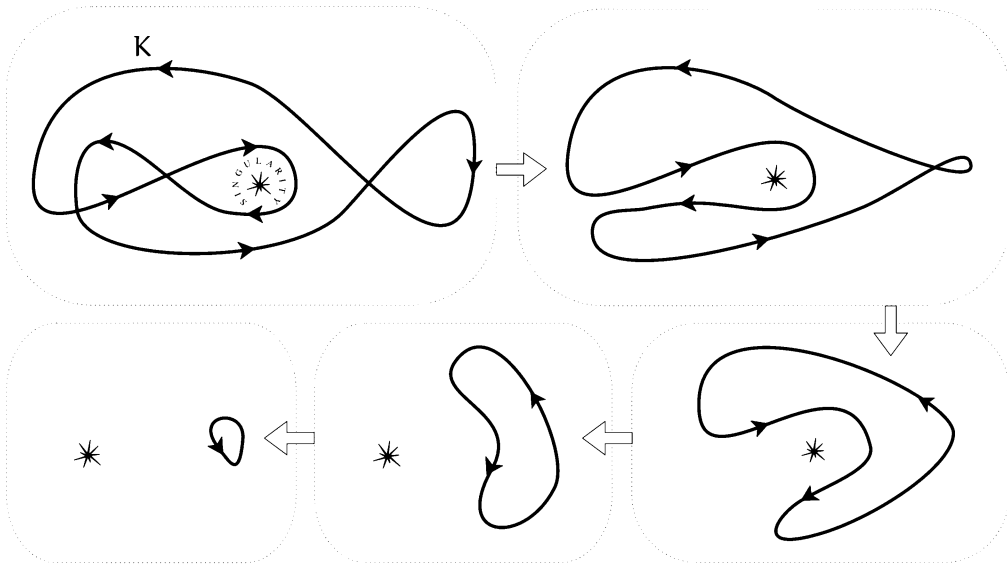
From the above example, it is clear that to prove the general version of Cauchy’s Theorem we need only show that (8.23) is true for *any* K .

Consider [8.33], which shows a portion of an arbitrary contour K sandwiched between two of the regions (D_j and D_k) into which it partitions the plane; also shown is the counterclockwise sense of their boundaries (C_j and C_k). Using the “crossing rule” (7.1) on page 388, we deduce that $\nu_j = \nu_k + 1$. Thus, in (8.23), we find that the contour C_j in the direction of K will always occur precisely one more time than the contour C_k in the opposite direction—the net result is that K is traversed once in the correct direction. Done.

As previously explained, in establishing (8.23) we have also deduced the General Cauchy Theorem (8.20).

8.11.3 A Simpler Explanation

The Deformation Theorem (8.11) was also deduced from the basic Cauchy theorem [unproven at that time], and we will now use it to give a simpler and more intuitive explanation of the General Cauchy Theorem.



[8.34] If a closed contour can be shrunk to a point without crossing a singularity, the integral round it vanishes. Hopf's Degree Theorem tells us that this shrinking process is possible if and only if the contour does not wind around any singularities.

Suppose that a contour can be deformed and shrunk down to a point without ever crossing a singularity of an otherwise analytic function. By inequality (8.5), the value of the integral will be zero at the end of this shrinking process. But by the Deformation Theorem, the value of the integral remains constant throughout this process. In other words,

If a closed contour can be shrunk to a point without crossing a singularity, the integral round it vanishes. (8.24)

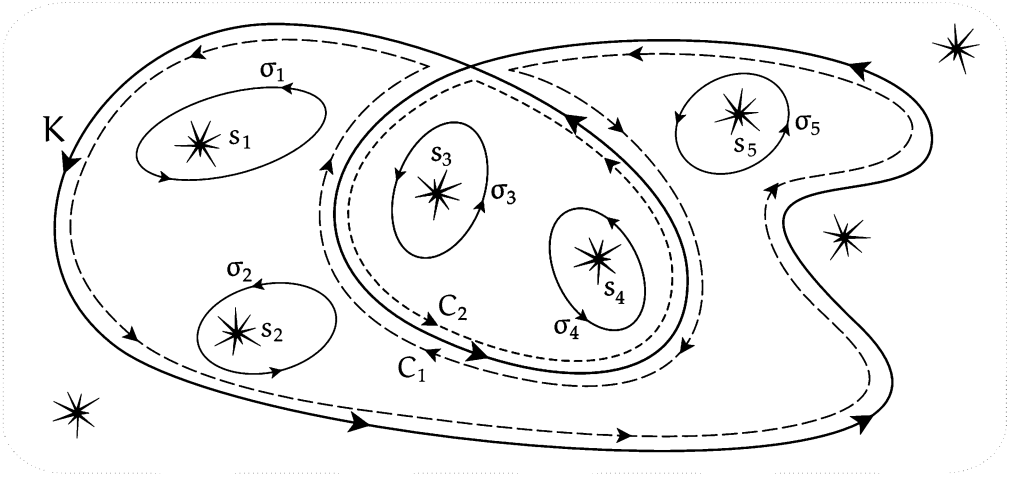
To wrap this up, we clearly need a way of recognizing when this shrinking process is possible. For example, is it possible for the contour K in [8.31]? Figure [8.34] shows that it is. Therefore (8.24) implies that the integral along K vanishes, in agreement with the general theorem.

The two theorems are clearly very closely related. In fact we can now deduce the General Cauchy Theorem from (8.24) by observing that the winding number of the final shrunken loop vanishes; Hopf's Degree Theorem then tells us that

The shrinking process in (8.24) is possible if and only if the contour does not wind around any singularities.

8.12 The General Formula of Contour Integration

Consider the general problem of evaluating $\oint_K f(z) dz$, where K is a general (possibly self-intersecting) loop, and where f possesses several singularities s_1, s_2 , etc., inside K . Figure [8.35] illustrates such a situation.



[8.35] The General Residue Theorem: $\oint_K f(z) dz = 2\pi i \sum_j \nu(K, s_j) \text{Res}[f(z), s_j]$. We know that $K = \sum_j \nu_j C_j$, and therefore $\oint_K f(z) dz = \sum_j \nu_j \oint_{C_j} f(z) dz$. The Deformation Theorem then tells us that each $\oint_{C_j} f(z) dz$ can be expressed as a sum of integrals around the illustrated loops σ_j encircling the singularities within each C_j . Finally, each such integral $\oint_{\sigma_k} f(z) dz = 2\pi i \text{Res}[f(z), s_k]$.

Here the inside of K consists of two simply connected regions, D_1 [lightly shaded] and D_2 [darkly shaded], with boundaries C_1 and C_2 . Since K winds once round points in the lightly shaded region ($\nu_1 = 1$) and twice round points in the darkly shaded region ($\nu_2 = 2$), the general result (8.23) correctly predicts that

$$K = \sum_j \nu_j C_j = 1 \cdot C_1 + 2 \cdot C_2. \quad (8.25)$$

As illustrated, let σ_j be a simple (counterclockwise) contour containing s_j but no other singularities of f , and let us define

$$I_j \equiv \oint_{\sigma_j} f(z) dz.$$

By virtue of the Deformation Theorem (8.11), we know that the integral I_j has a characteristic value that does not depend on the size or shape of σ_j . Furthermore, as we saw in [8.20], if a simple loop contains several singularities, then the integral round that loop is the sum of I -values of the singularities it contains. In our example,

$$\oint_{C_1} f(z) dz = I_1 + I_2 + I_5 \quad \text{and} \quad \oint_{C_2} f(z) dz = I_3 + I_4.$$

Finally, using (8.25), we deduce that

$$\oint_K f(z) dz = 1 \cdot [I_1 + I_2 + I_5] + 2 \cdot [I_3 + I_4],$$

in which *each* I -value has been multiplied by the number of times K winds round the corresponding singularity. Since (8.23) is valid for arbitrary loops it follows that this conclusion is too. As the grand finale to this chapter, we have thus obtained the following completely general formula for the loop integral of an analytic function:

$$\oint_K f(z) dz = \sum_j v(K, s_j) I_j.$$

The final icing on the cake is an efficient method of computing the I_j 's. In the next chapter we will verify our previous claim that in the neighbourhood of each s_j there exists a unique Laurent series [see page 456], the coefficient of the complex inversion term being (by definition) the residue $\text{Res}[f(z), s_j]$. Granted this, we see that $I_j = 2\pi i \text{Res}[f(z), s_j]$. Thus

$$\oint_K f(z) dz = 2\pi i \sum_j v(K, s_j) \text{Res}[f(z), s_j]. \quad (8.26)$$

This is the *General Residue Theorem*. Note that it contains the General Cauchy Theorem as the special case in which each $v(K, s_j) = 0$.

We will also see in the next chapter that it is possible to find the residues in this formula directly, without going to the trouble of finding the whole Laurent series. Thus, even before exemplifying its use, it should be clear that in (8.26) we have a result of great practical and theoretical power.

8.13 Exercises

- 1 Thinking of x as representing time, $z(x) = x + if(x)$ is a parametric description of the ordinary graph, $y = f(x)$.

(i) Show that the complex velocity is $v = 1 + i \tan \theta$, where θ is the angle between the horizontal and the tangent to the graph. Also, show that complex acceleration is $a = i f''$.

- (ii) Recall from Ex. 20 on page 297 that the curvature of the orbit is $\kappa = [\text{Im}(a \bar{v})]/|v|^3$. Deduce from (i) that

$$\kappa \sec^3 \theta = f''(x).$$

- (iii) From (ii), deduce that the error equation (8.3) can be written as

$$\text{area}(ABCD) = \frac{1}{8} f''(x) \Delta^3.$$

- 2 In [8.6], show that

$$\lim_{\Delta \rightarrow 0} \left(\frac{\text{area between the chord AB and the curve}}{\text{area between the tangent CD and the curve}} \right) = 2.$$

In other words, R_M is twice as accurate as the Trapezoidal formula.

- 3 In the integration of an ordinary real function $f(x)$, let L denote the length of the integration range, and let M denote the maximum size of $f''(x)$ in this range. From the previous two exercises, deduce the standard result,

$$\text{total Trapezoidal error} < \frac{1}{12} LM \Delta^2.$$

Likewise, deduce the somewhat less familiar result,

$$\text{total } R_M \text{ error} < \frac{1}{24} LM \Delta^2.$$

- 4 Write down the values of $\oint_C (1/z) dz$ for each of the following choices of C , then confirm the answers the hard way, using parametric evaluation.

(i) $|z| = 1$.

(ii) $|z - 2| = 1$.

(iii) $|z - 1| = 2$.

- 5 Evaluate parametrically the integral of $(1/z)$ round the square with vertices $\pm 1 \pm i$, and confirm that the answer is indeed $2\pi i$.

- 6 Confirm by parametric evaluation that the integral of z^m round an origin-centred circle vanishes, except when $m = -1$.

- 7 Hold a coin (of radius A) down on a flat surface and roll another one (of radius B) round it. The path traced by a point on the rim of the rolling coin is called an *epicycloid*, and it is a closed curve if $A = nB$, where n is an integer.

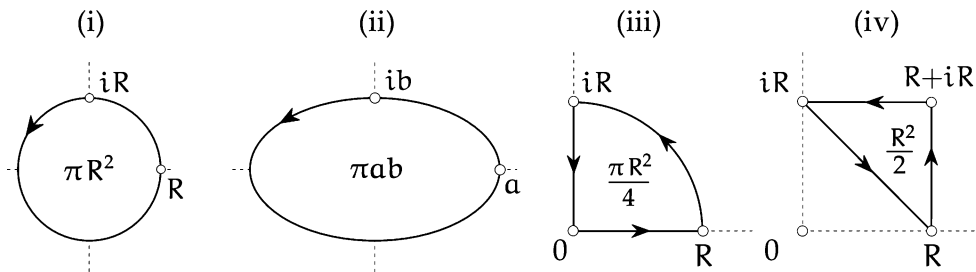
- (i) With the centre of the fixed coin at the origin, show that the epicycloid can be represented parametrically as

$$z(t) = B \left[(n+1) e^{it} - e^{i(n+1)t} \right].$$

- (ii) By evaluating the integral in (8.8) parametrically, show that

$$\text{area of epicycloid} = \pi B^2 (n+1)(n+2).$$

- 8 The figure below shows four simple loops, and in each case we have indicated how much shaded area is enclosed. Use parametric evaluation to verify equation (8.8) for each of the four loops.



- 9 What is the generalization of (8.8) to the case where the contour is not closed?
- 10 Use (8.23) to verify (8.9).
- 11 The perfect symmetry of figure [8.18] results from integration round the *unit* circle. Roughly how would this figure look if we instead used a somewhat larger circle?
- 12 Let K be the contour in [8.21].
- (i) Evaluate the following integral by factoring the denominator and putting the integrand into partial fractions:

$$\oint_K \left(\frac{z}{z^2 - iz - 1 - i} \right) dz.$$

- (ii) Write down the Laurent series (centred at the origin) for $(\cos z/z^{11})$. Hence find

$$\oint_K \left(\frac{\cos z}{z^{11}} \right) dz.$$

- 13 This exercise illustrates how one type of difficult real integral may be evaluated easily using a complex integral.

Let L be the straight contour along the real axis from $-R$ to $+R$, and let J be the semi-circular contour (in the upper half plane) back from $+R$ to $-R$. The complete contour $L + J$ is thus a closed loop.

- (i) Using the partial fraction idea of the previous exercise, show that the integral

$$\oint_{L+J} \frac{dz}{(z^4 + 1)}$$

vanishes if $R < 1$, and find its value if $R > 1$.

- (ii) Using the fact that $z^4 + 1$ is the complex number from -1 to z^4 , write down the minimum value of $|z^4 + 1|$ as z travels round J . Now think of R as large, and use inequality (8.5) to show that the integral round J dies away to zero as R grows to infinity.
- (iii) From the previous parts, deduce the value of

$$\int_{-\infty}^{+\infty} \frac{dx}{(x^4 + 1)}.$$

- 14 (i) The integral

$$\int_{-\infty}^{+\infty} \frac{dx}{(x^2 + 1)}$$

is easily found by ordinary means, but evaluate it instead by the method of the previous exercise.

- (ii) Likewise, evaluate

$$\int_{-\infty}^{+\infty} \frac{dx}{(x^2 + 1)^2}$$

by ordinary means and then by contour integration. [*Hint:* The quickest way to find the partial fraction decomposition for this function is to square the decomposition of $1/(z^2 + 1)$.]

- 15 (i) Use the Fundamental Theorem to write down the value of

$$\int_0^{a+ib} e^z dz.$$

- (ii) Equate the answer with the one obtained by parametric evaluation along the straight contour from 0 to $(a + ib)$, and deduce that

$$\int_0^1 e^{ax} \cos bx \, dx = \frac{a(e^a \cos b - 1) + b e^a \sin b}{a^2 + b^2},$$

and

$$\int_0^1 e^{ax} \sin bx \, dx = \frac{b(1 - e^a \cos b) + a e^a \sin b}{a^2 + b^2}.$$

- (iii) Prove the results in (ii) by ordinary methods.

- 16 (i) Show that when integrating a product of analytic functions, we may use the ordinary method of “integration by parts”.
 (ii) Let L be a contour from the real number $-\theta$ to $+\theta$. Show that

$$\int_L z e^{iz} dz = 2i (\sin \theta - \theta \cos \theta),$$

and verify this by taking L to be a line-segment and integrating parametrically.

17 Let

$$f(z) = \frac{1}{z} \left(z + \frac{1}{z} \right)^n,$$

where n is a positive integer.

- (i) Use the Binomial Theorem to find the residue of f at the origin when n is even and when n is odd.
 (ii) If n is odd, what is the value of the integral of f round any loop?
 (iii) If $n = 2m$ is even and C is a simple loop winding once round the origin, deduce from part (i) that

$$\oint_C f(z) dz = 2\pi i \frac{(2m)!}{(m!)^2}.$$

- (iv) By taking C to be the unit circle, deduce the following result due to Wallis:

$$\int_0^{2\pi} \cos^{2m} \theta d\theta = \frac{(2m)!}{2^{2m-1} (m!)^2} \pi.$$

- (v) Similarly, by considering functions of the form $z^k f(z)$ where k is an integer, evaluate

$$\int_0^{2\pi} \cos^n \theta \cdot \cos k\theta d\theta \quad \text{and} \quad \int_0^{2\pi} \cos^n \theta \cdot \sin k\theta d\theta$$

- 18 Let E be the elliptical orbit $z(t) = a \cos t + ib \sin t$, where a and b are positive and t varies from 0 to 2π . By considering the integral of $(1/z)$ round E , show that

$$\int_0^{2\pi} \frac{dt}{a^2 \cos^2 t + b^2 \sin^2 t} = \frac{2\pi}{ab}.$$

- 19 Let us verify the claim of Ex. 19, p. 297, that if a function has vanishing Schwarzian derivative, then it must be a Möbius transformation. Following Beardon (1984), suppose that $\{f(z), z\} = 0$, and define $F \equiv (f''/f')$.

- (i) Show that $1/F(z) - 1/F(w) = -(z - w)/2$.
 (ii) Deduce that $\frac{d}{dz} \log f'(z) = -2/(z - a)$, for some constant a .
 (iii) Perform two further integrations to conclude that $f(z)$ is a Möbius transformation.

- 20 In [8.27], consider the white fragments of squares sandwiched between K and C .
- (i) Show that the sum of the integrals round these fragments equals the difference between the integrals round C and K .
 - (ii) As ϵ shrinks, what is the approximate size of each term in the above series?
 - (iii) Roughly how many terms are there in the series?
 - (iv) From the previous parts, what do you conclude about the difference between the integrals round C and K , as ϵ shrinks to nothing?
- 21 Let K be the straight contour from $a - (\epsilon/2)$ to $a + (\epsilon/2)$, where ϵ is a short complex number in an arbitrary direction.
- (i) Use the Fundamental Theorem to integrate z^2 along K , and then write down the value obtained by using a single term in R_M . Show that the error induced by R_M is $\frac{1}{12} \epsilon^3$.
 - (ii) As in part (i), find both the exact value and the R_M value for the integral of e^z along K . By expanding $e^{\epsilon/2}$ as a power series, deduce that the error in this case is roughly $\frac{1}{24} e^a \epsilon^3$.
 - (iii) Repeat the error analysis of the previous parts for the non-analytic function $\bar{z}^2$. [You will need to use parametric evaluation to find the exact value of the integral.]
- 22 Let K be the short contour of the previous exercise. Suppose that $f(z)$ possesses a Taylor series centred at a that converges at points of K :

$$f(a+h) = f(a) + \frac{f'(a)}{1!} h + \frac{f''(a)}{2!} h^2 + \frac{f'''(a)}{3!} h^3 + \dots$$

[The existence of such a series for any analytic function is derived in the next chapter.]

- (i) By integrating this series along K , show that the difference between the exact integral and the R_M value is roughly $\frac{1}{24} f''(a) \epsilon^3$. Verify that the results of the first two parts of the previous exercise are in accord with this finding.
- (ii) Use the series to show that the complex number from the image of the midpoint of K to the midpoint of the images of the ends of K is roughly $\frac{1}{4} f''(a) \epsilon^2$. As ϵ shrinks, are these two types of midpoint distinguishable under the magnifying lens that produces figure [8.28]?
- (iii) From the Fundamental Theorem, deduce that the existence of such a series implies the vanishing of the integral of f round loops within the disc of convergence.

- 23 Let $f(z)$ be analytic throughout a region which contains a triangle with vertices a, b, c , and hence with edges $A \equiv (c - b)$, $B \equiv (a - c)$, $C \equiv (b - a)$. Given a pair of point p and q , let us define w_{pq} as a kind of average of $f(z)$ along the line-segment pq :

$$w_{pq} \equiv \frac{1}{(q - p)} \int_p^q f(z) dz.$$

Show that this complex average mapping sends the sides of the triangle abc to the vertices w_{ab}, w_{bc}, w_{ca} of a *similar* triangle! We merely rediscovered this result, which is apparently due to [Echols \(1923\)](#).

[Hint: Show that $Aw_{bc} + Bw_{ca} + Cw_{ab} = 0$, and use $A + B + C = 0$.]

- 24 Let K be a closed contour, and let ν be its winding number about the point a . Show that

$$\oint_K \left(\frac{e^z}{z - a} \right) dz = 2\pi i \nu e^a.$$

[Hint: Write e^z as $e^a e^{(z-a)}$, and expand $e^{(z-a)}$ as a power series.] This is a special case of *Cauchy's Integral Formula* (explained in the next chapter), which states that if f is analytic inside K , then

$$\oint_K \frac{f(z)}{(z - a)} dz = 2\pi i \nu f(a).$$

- 25 Consider the image of the disc $|z| \leq R$ under the mapping $z \mapsto kz^m$. As the radius sweeps round the disc once, its image sweeps m times round the image disc of radius $|k|R^m$. Thus we may sensibly define the area of the image to be $m\pi(|k|R^m)^2$. With this understanding, show that if a mapping has a convergent power series

$$f(z) = a + bz + cz^2 + dz^3 + \dots,$$

then the area of the image is just the sum of the areas of the images under each of the separate terms of the series:

$$\text{area of image} = \pi(|b|^2 R^2 + 2|c|^2 R^4 + 3|d|^2 R^6 + \dots).$$

This is *Bieberbach's Area Theorem*.

Hint: Recall that the local area expansion factor is $|f'|^2$, so the image area is

$$\iint_{|z| \leq R} |f'|^2 dx dy = \int_0^R \left[\int_0^{2\pi} f'(r e^{i\theta}) \overline{f'(r e^{i\theta})} d\theta \right] r dr.$$

- 26 (i) Show that if f is an analytic function without singularities or p -points on a loop L , then

$$\nu[f(L), p] = \frac{1}{2\pi i} \oint_L \frac{f'(z)}{f(z) - p} dz.$$

(ii) Now let

$$f(z) = \frac{(z - a_1)^{A_1} (z - a_2)^{A_2} \cdots (z - a_n)^{A_n}}{(z - b_1)^{B_1} (z - b_2)^{B_2} \cdots (z - b_m)^{B_m}},$$

and by considering $(\log f)'$, find (f'/f) .

(iii) In part (i) put $p = 0$ and take L to be a simple loop containing the roots a_1 to a_r and containing the poles b_1 to b_s . Thereby obtain a calculation proof of the Generalized Argument Principle in the case of rational functions:

$$\begin{aligned} \nu[f(L), 0] &= \sum_{j=1}^r A_j - \sum_{j=1}^s B_j \\ &= (\text{number of interior roots}) - (\text{number of interior poles}). \end{aligned}$$